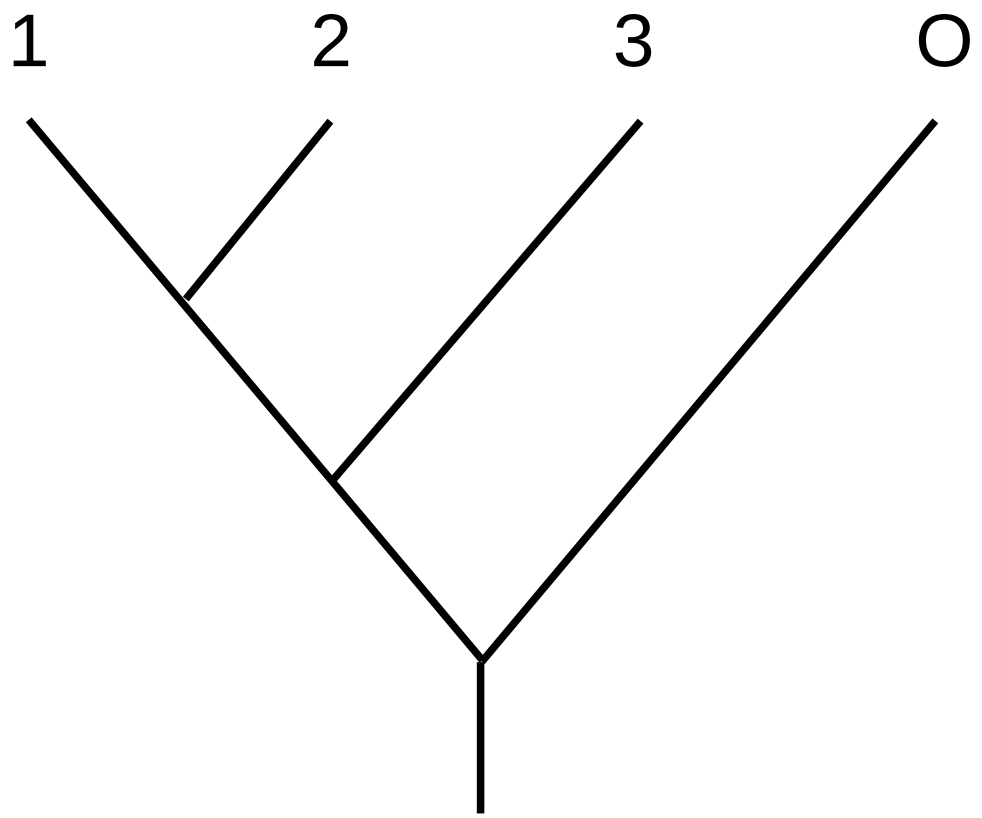
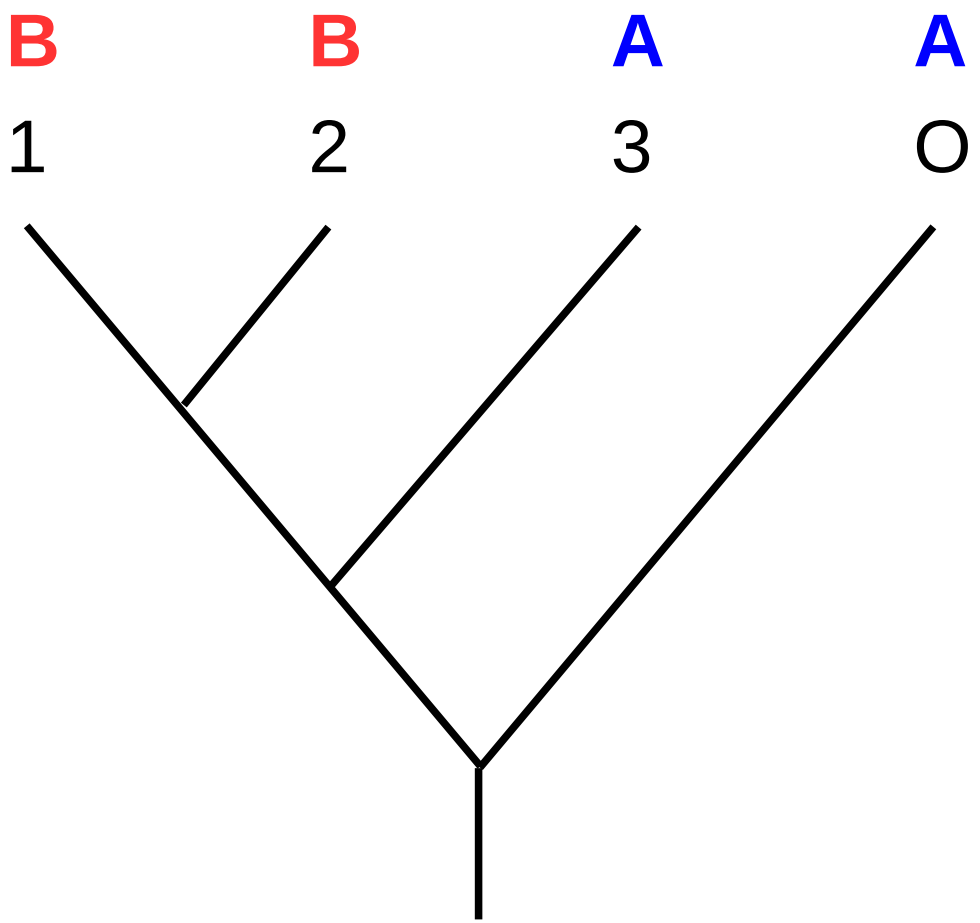
# Admixture: basic concepts and terminology

- Species and population
- ILS = incomplete lineage sorting. Retention of ancestral polymorphism through successive divergence (speciation) events
- Drift. Not all individuals reproduce equally. Leads to random change in allele frequencies in the subsequent generation. Can lead to allele extinction or fixation.
- Admixture. Genetic exchange between formerly isolated populations (gene flow, hybridisation)
- Ancestral and derived alleles. Ancestral inferred using outgroup, derived evolves from ancestral

# Species tree = the <u>true</u> population or speciation history

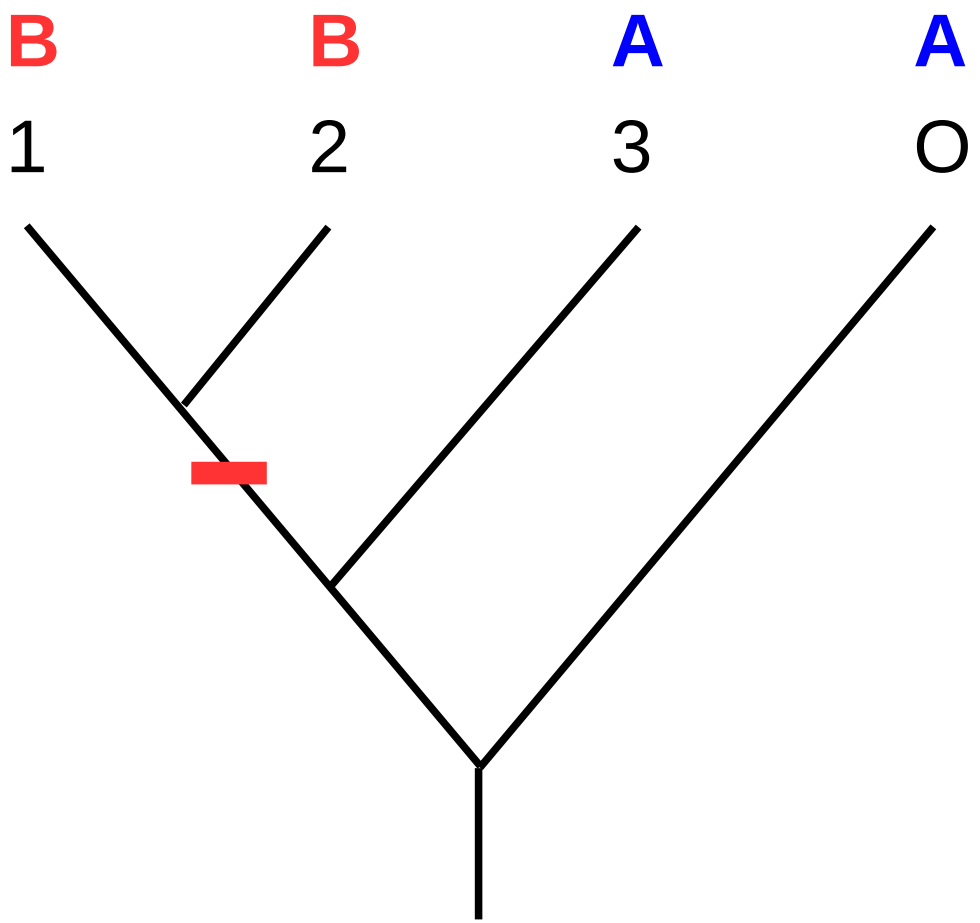





Biallelic locus

**A** = ancestral

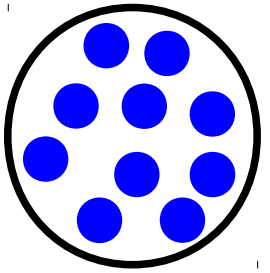
**B** = derived



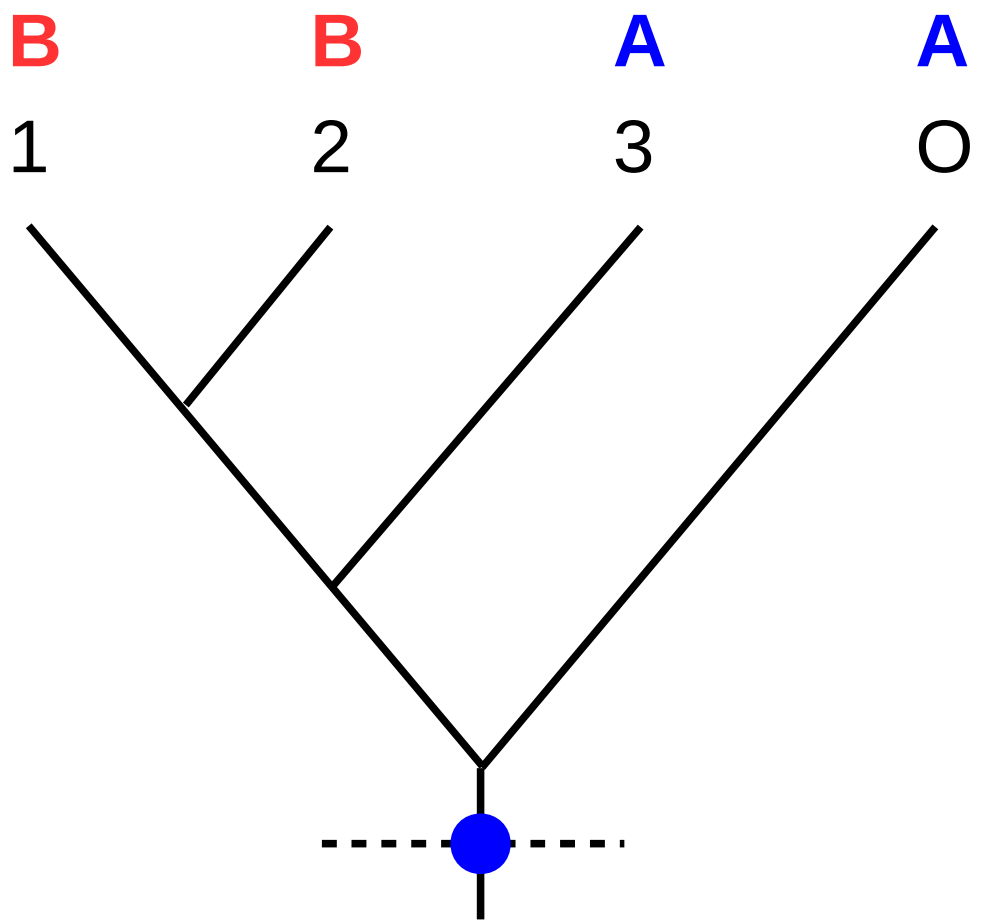

Biallelic locus

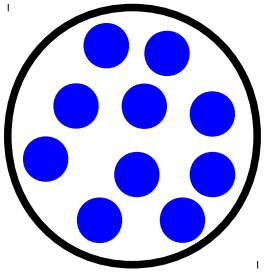
**A** = ancestral

**B** = derived

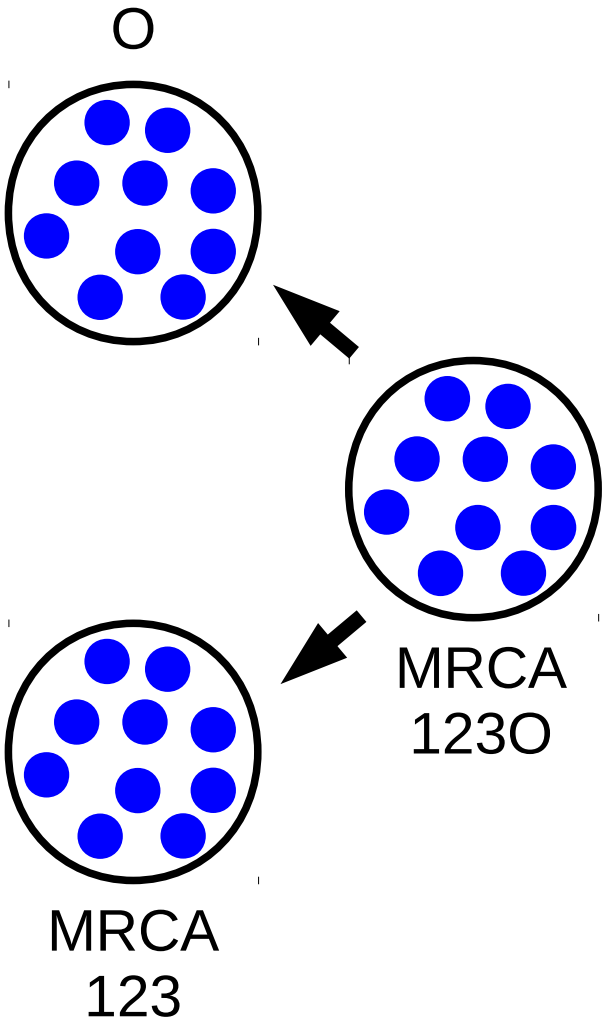


MRCA
123O

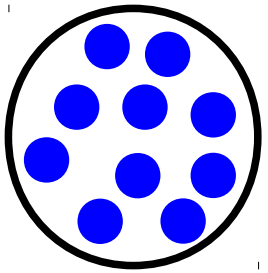

B
B
A
A
1
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3
O

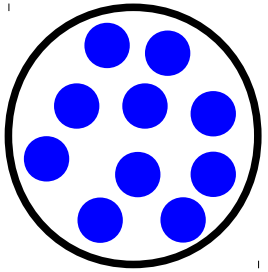


MRCA
123O

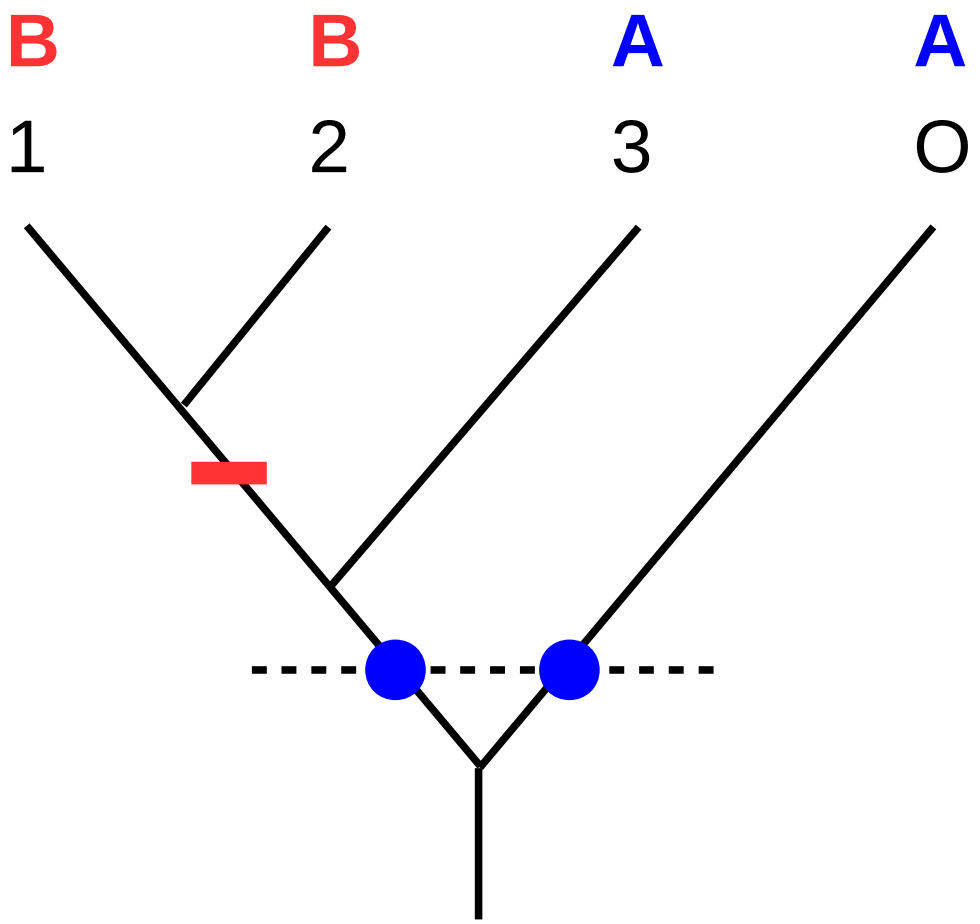

O
MRCA
123O
MRCA
123

O

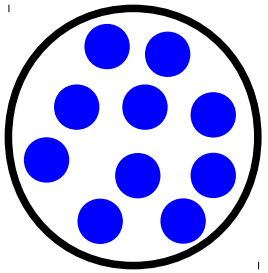


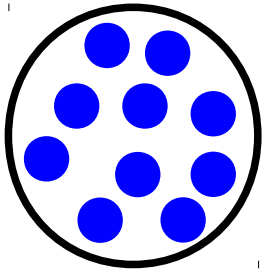


MRCA
123

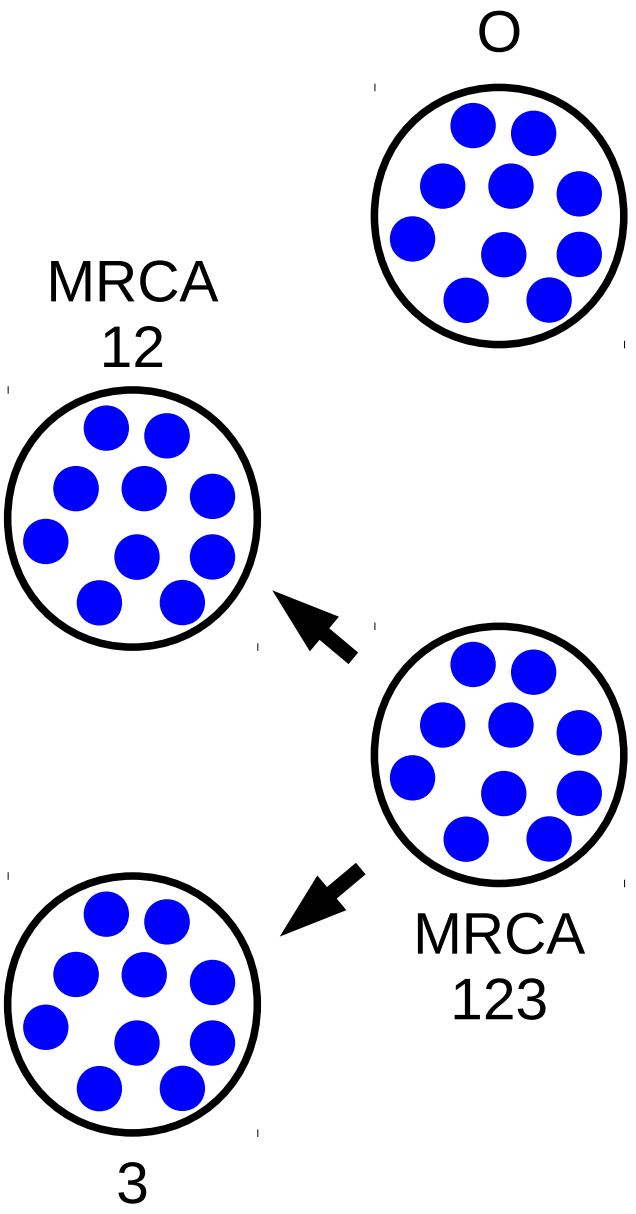

B
B
A
A
1
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O

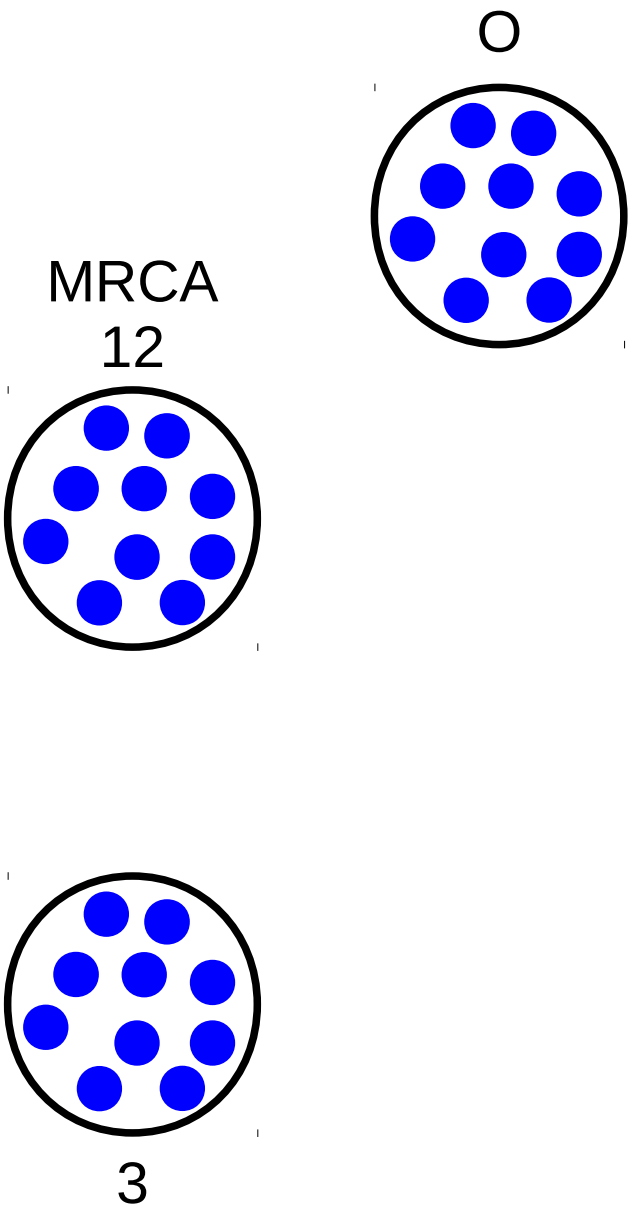
O

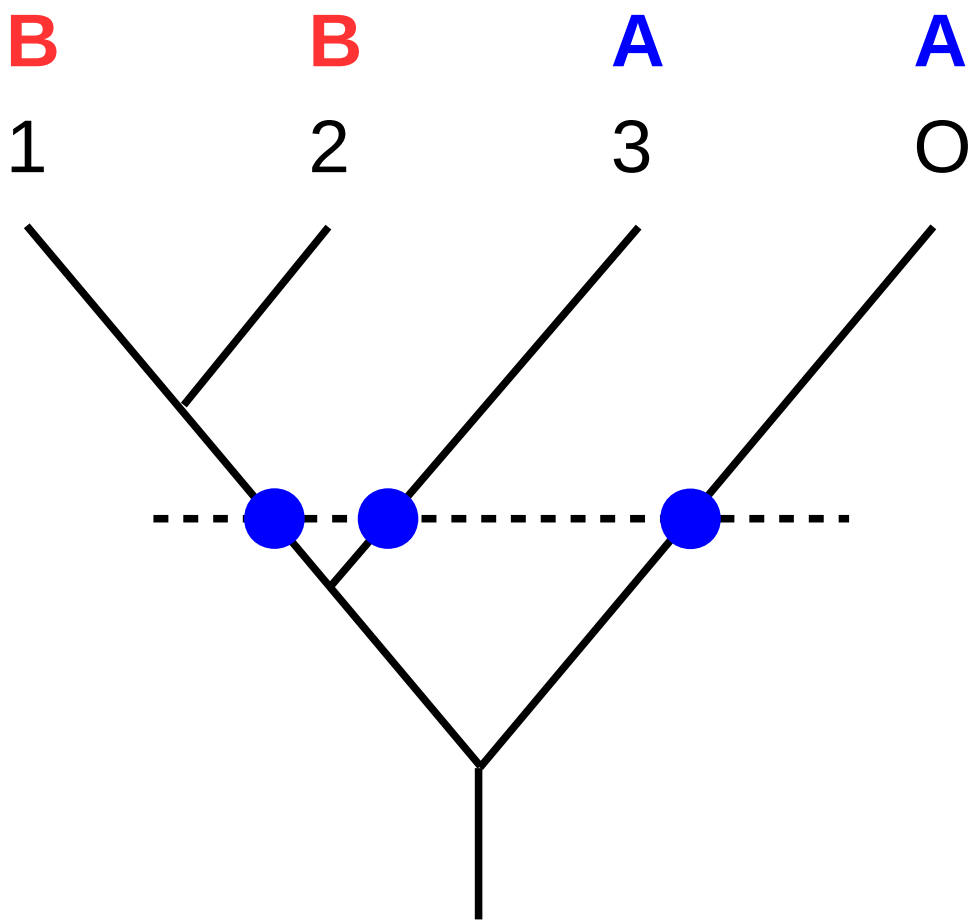


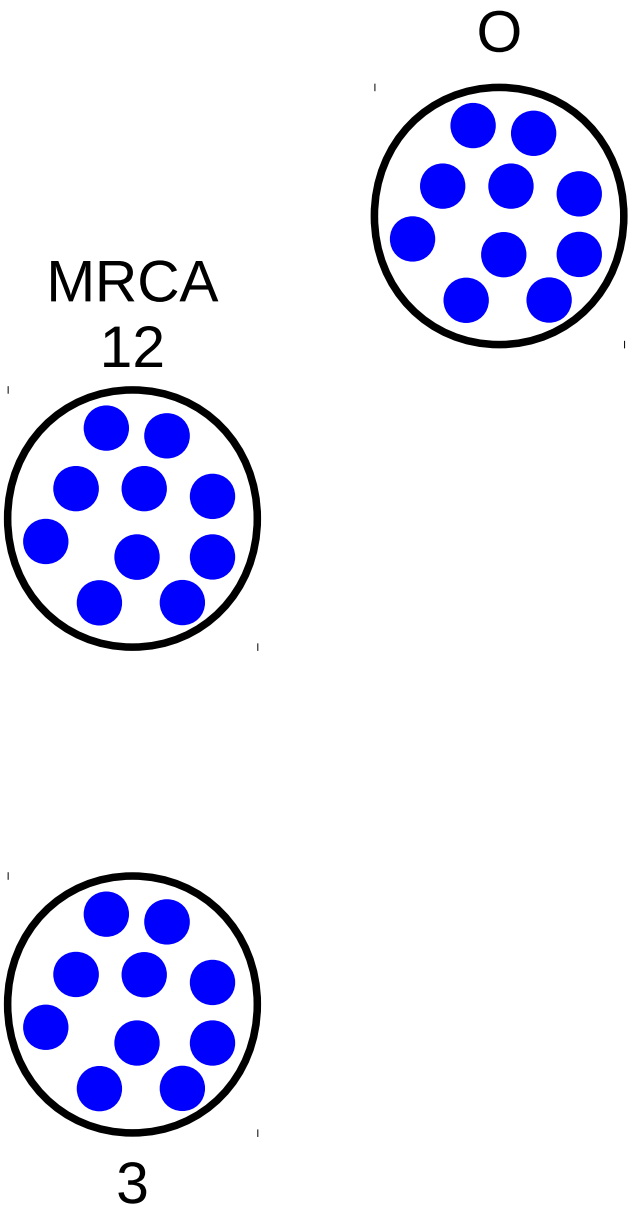


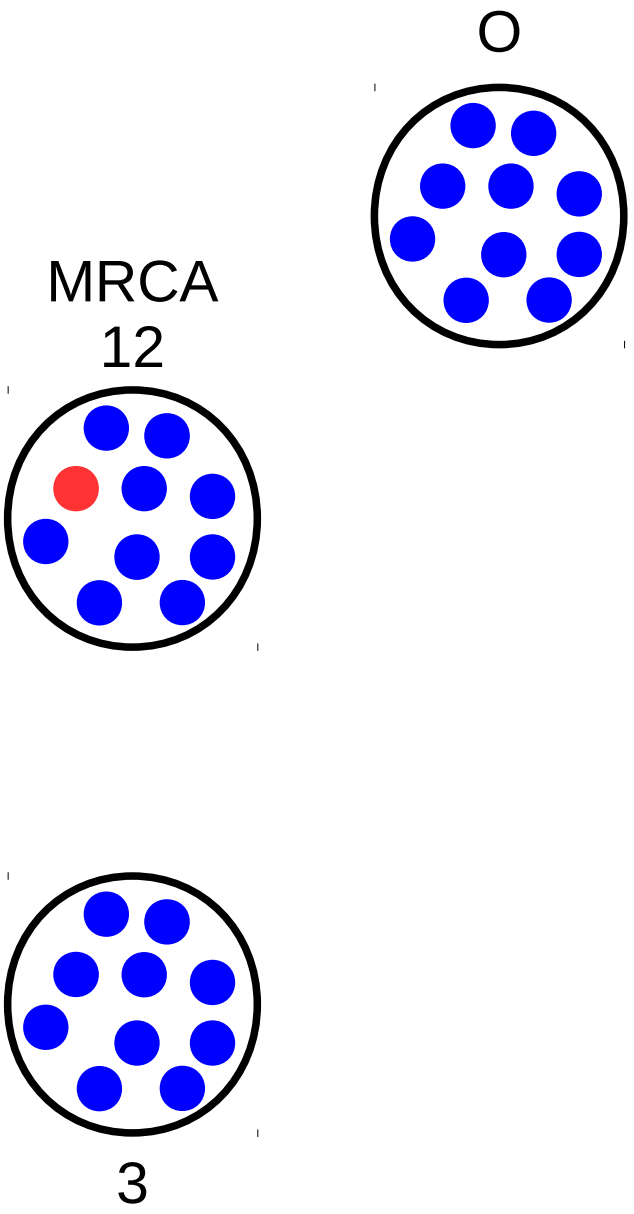
MRCA
123

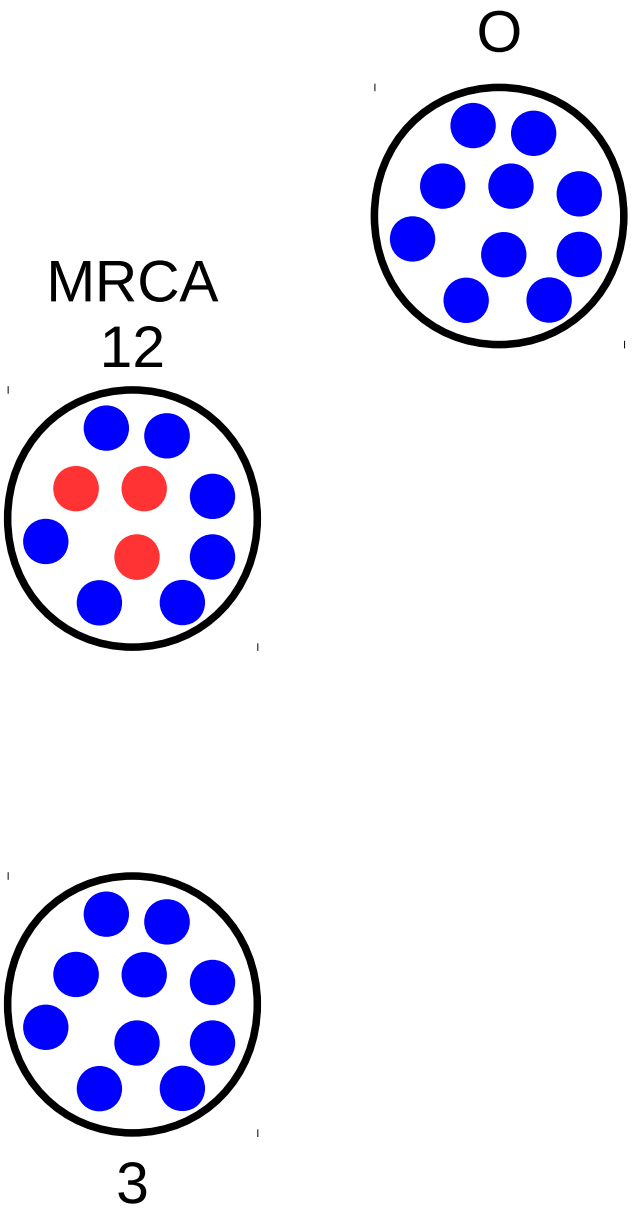

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MRCA
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MRCA
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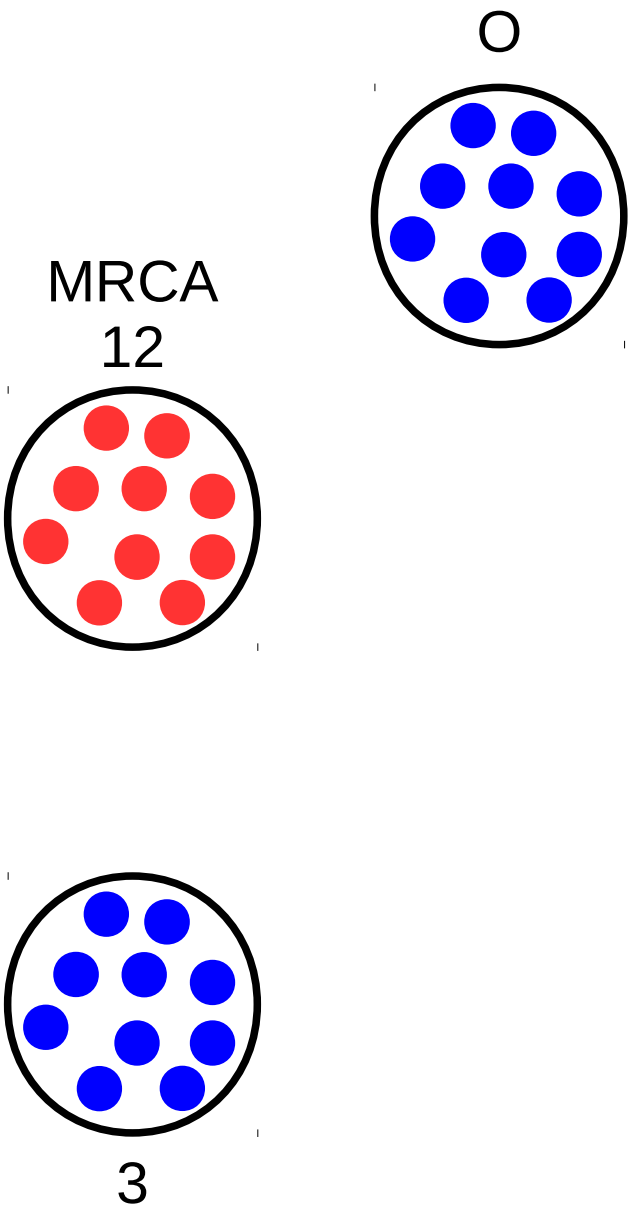

O
MRCA
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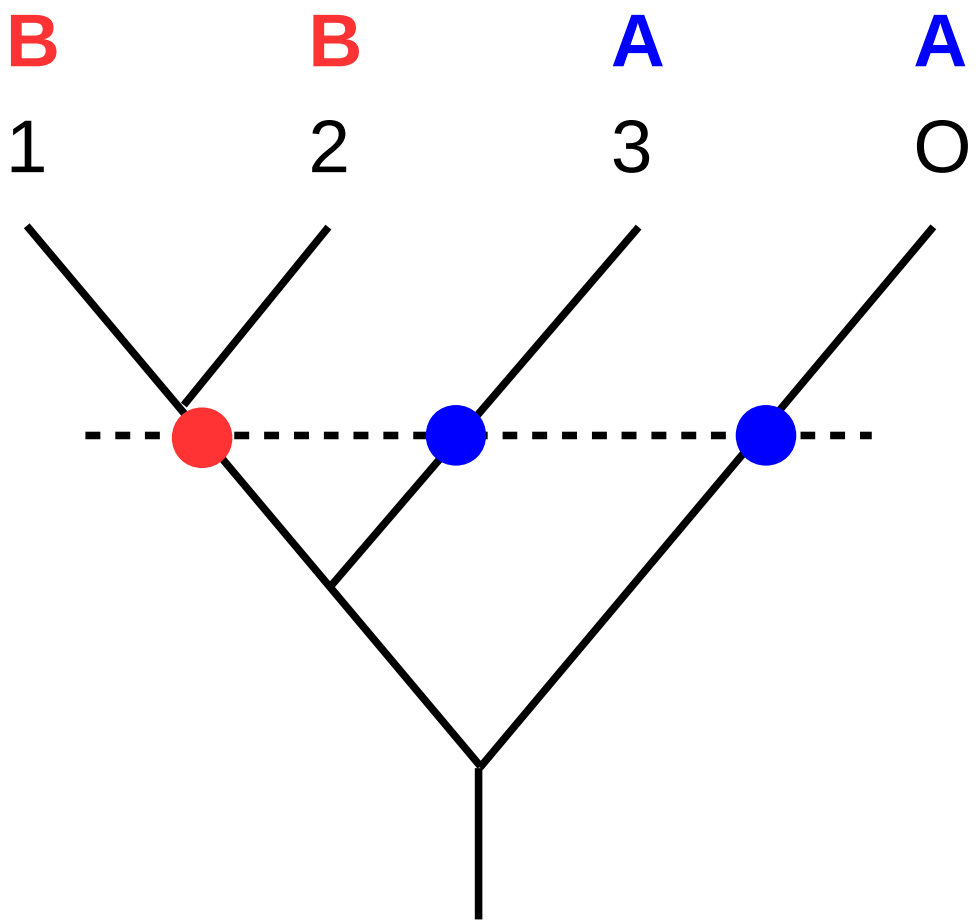

B
B
A
A
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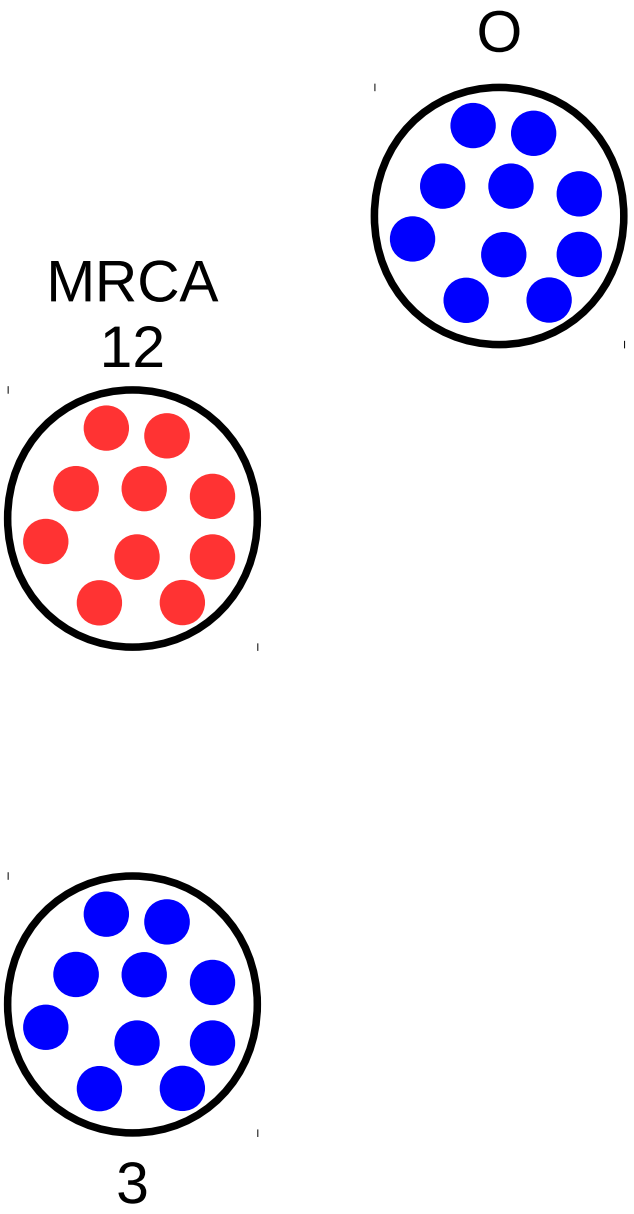

O
MRCA
12
3

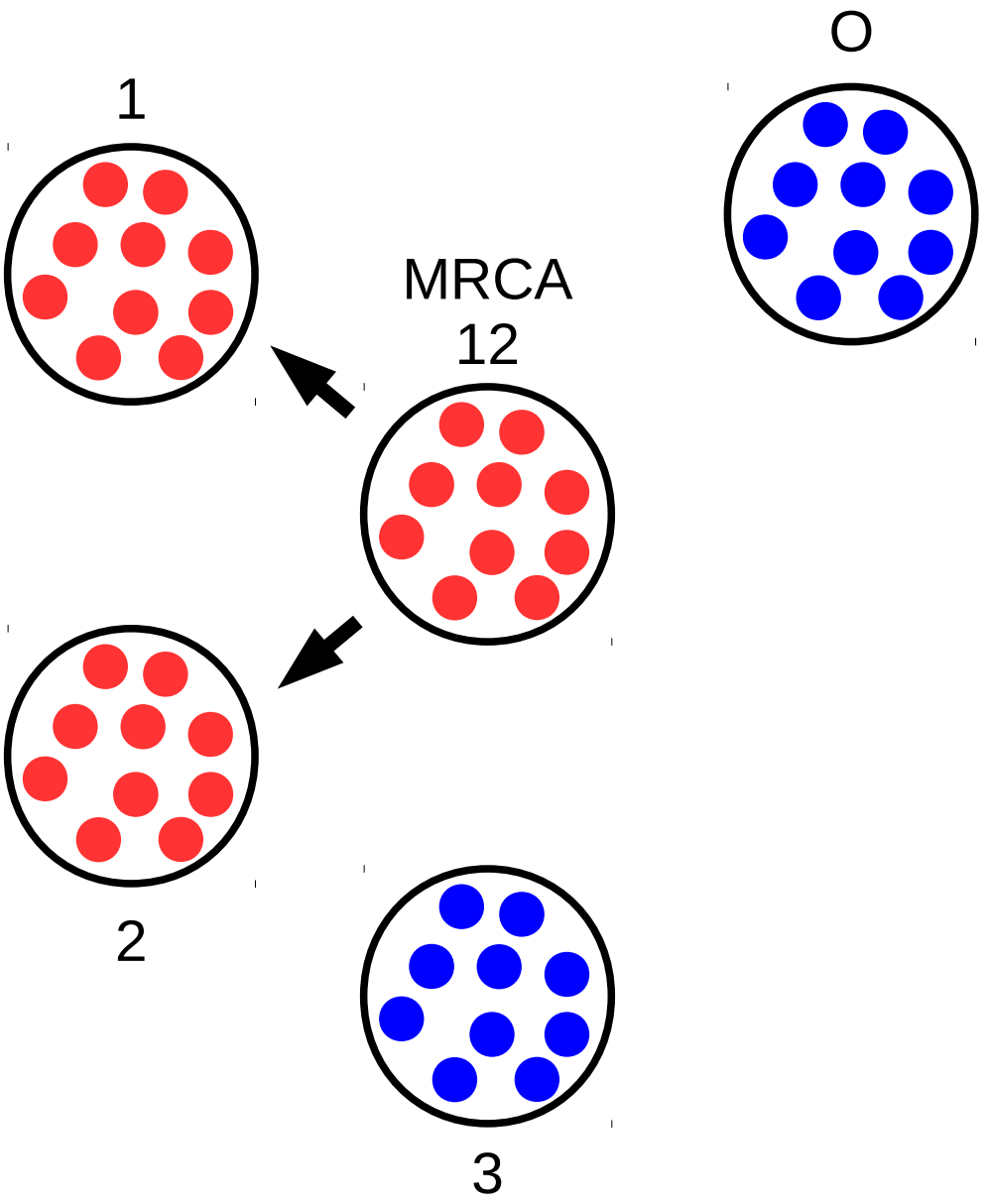

O
MRCA
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3

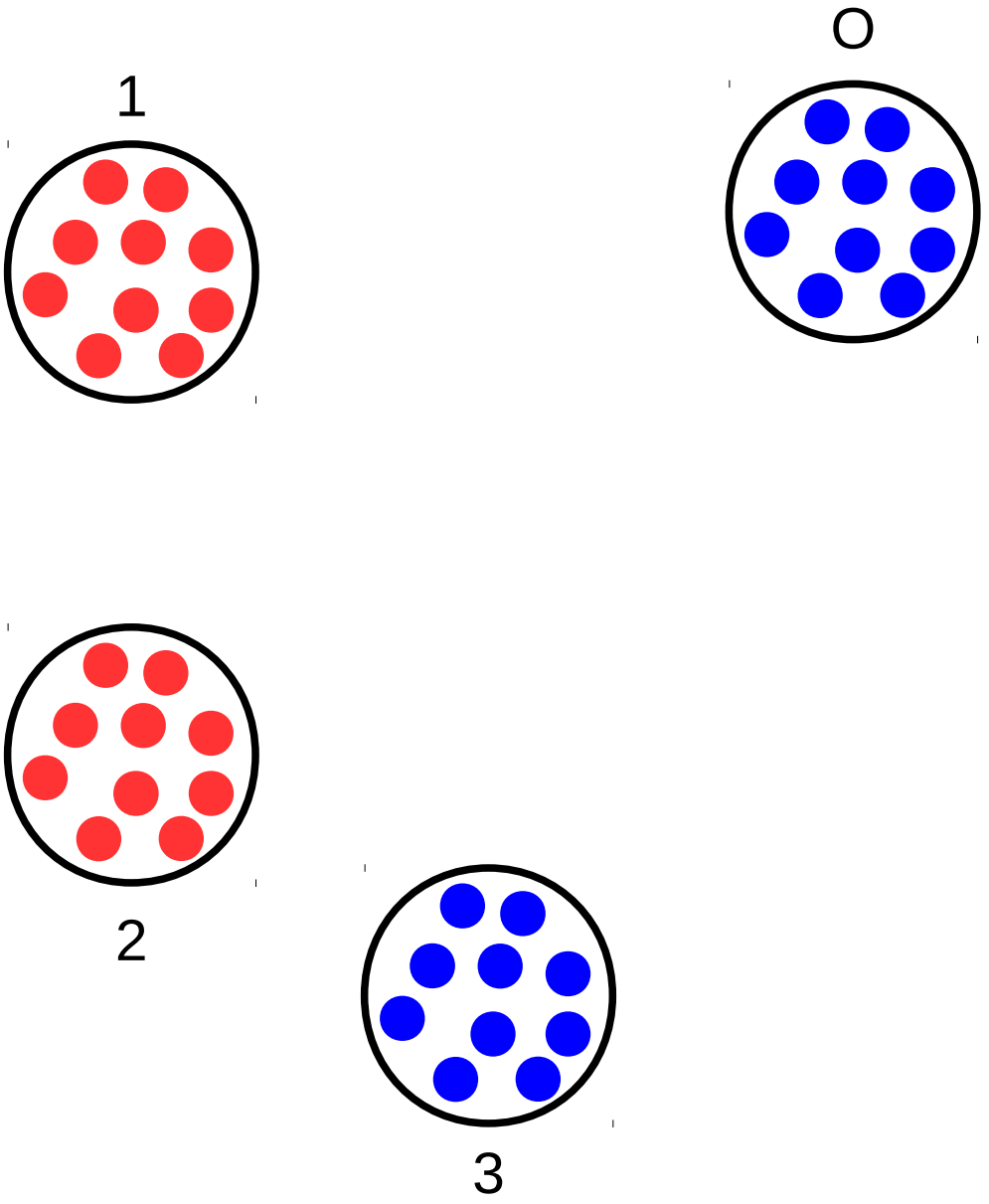

O
MRCA
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O
MRCA
12
3

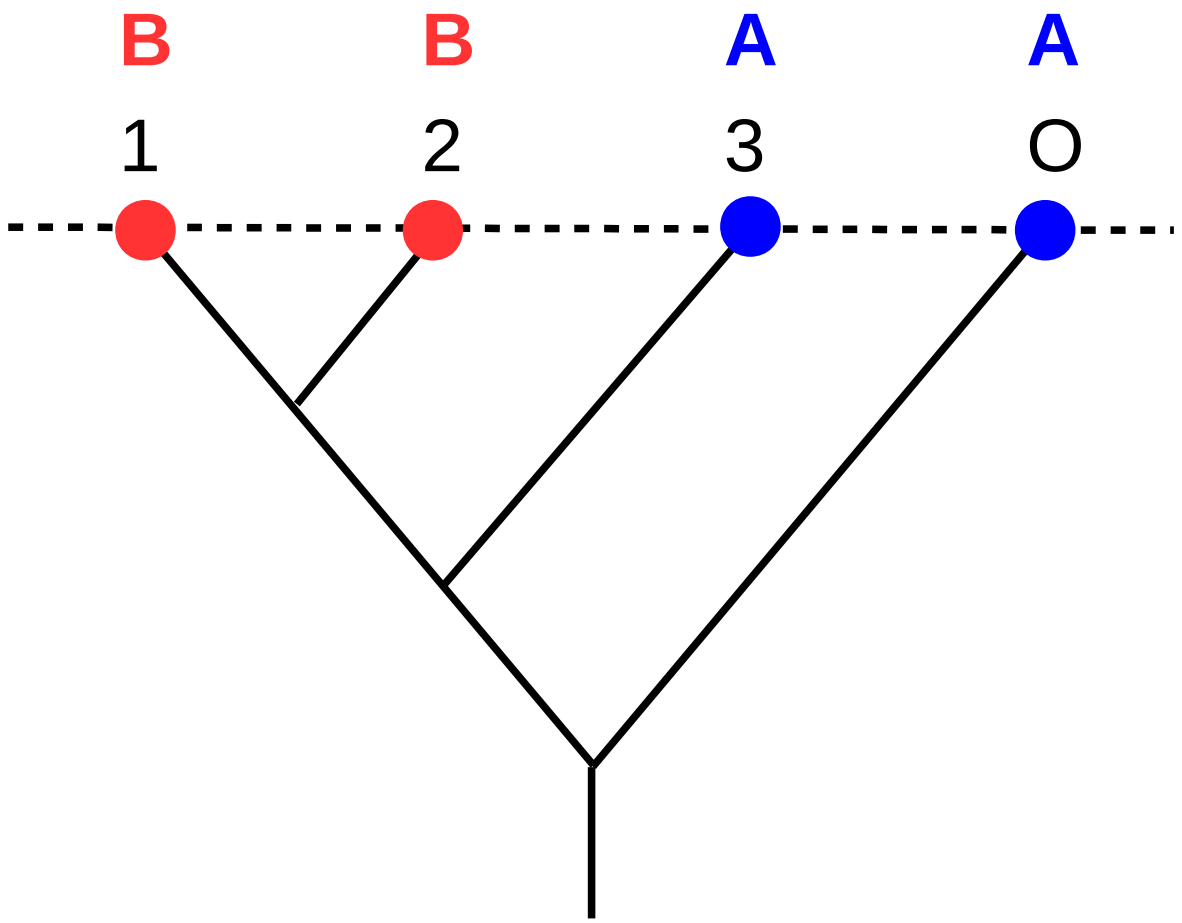

B
B
A
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O
MRCA
12
3

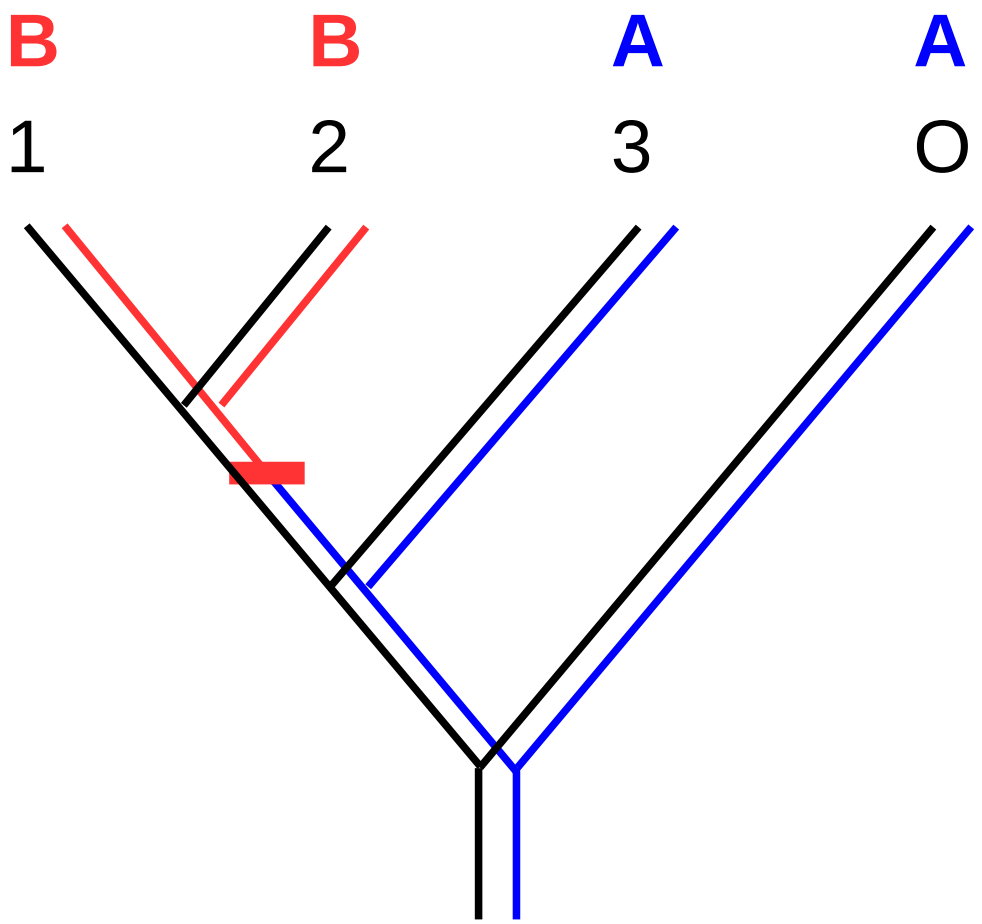

O
1
MRCA
12
2
3


1
O
2
3

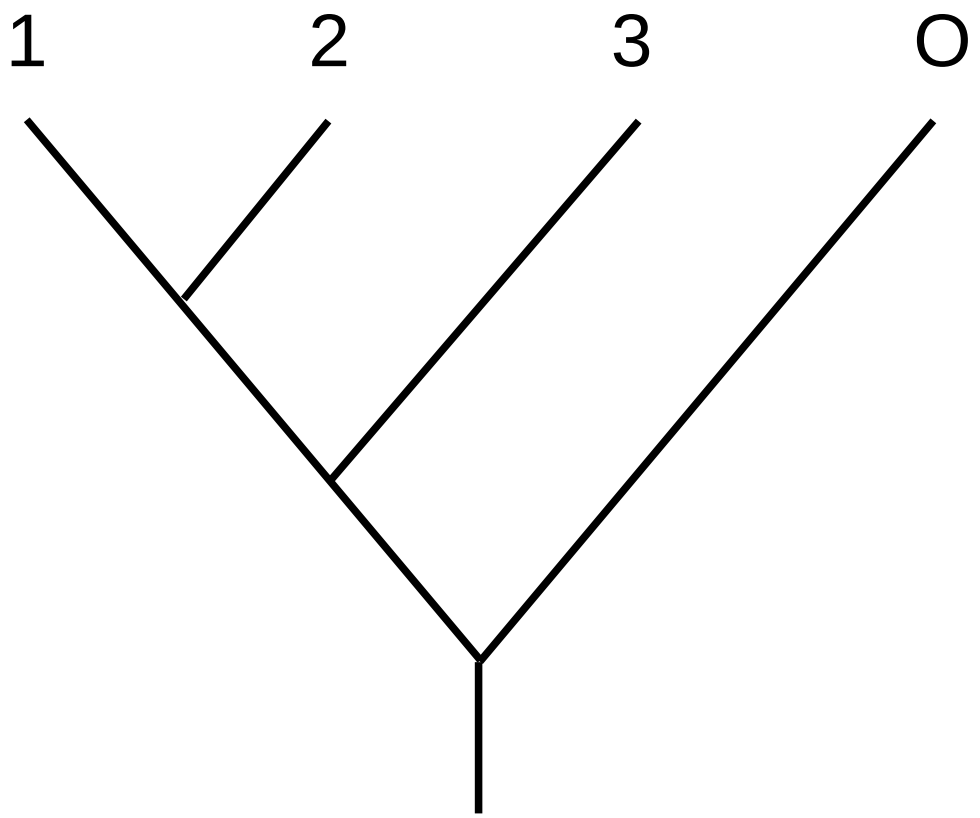
# Gene tree = species tree!



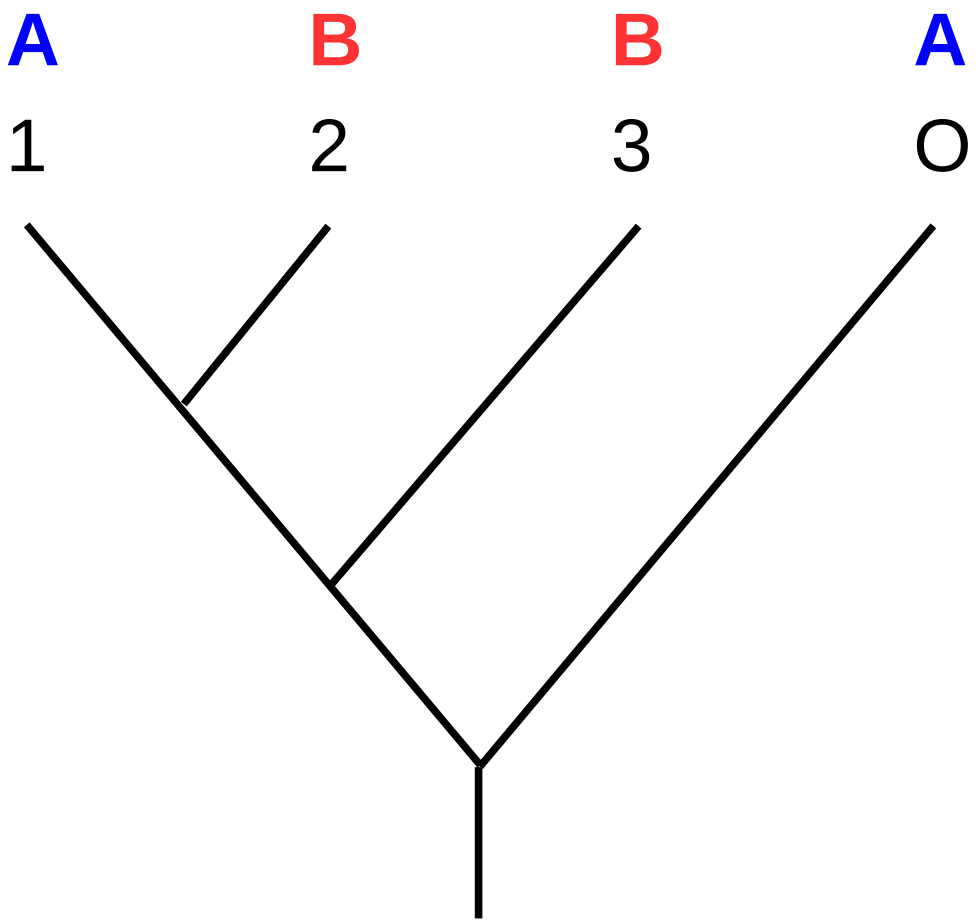
# Gene tree = species tree!



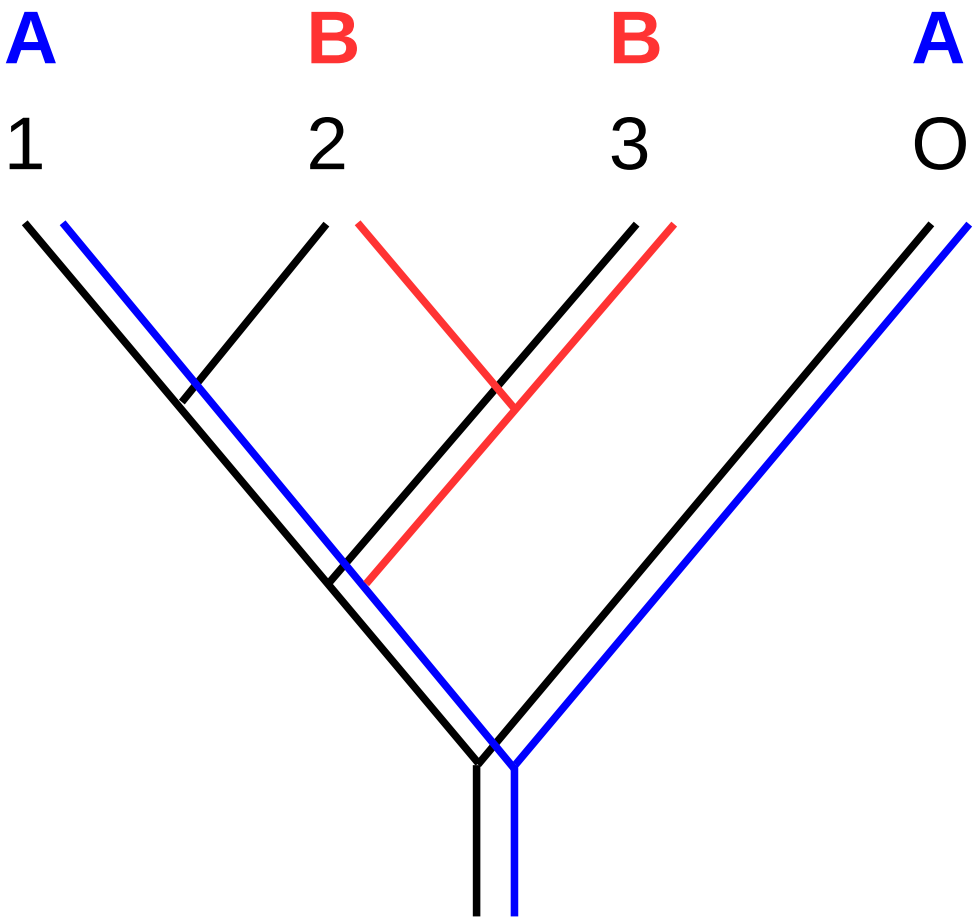
# BUT gene tree may not =  species tree!



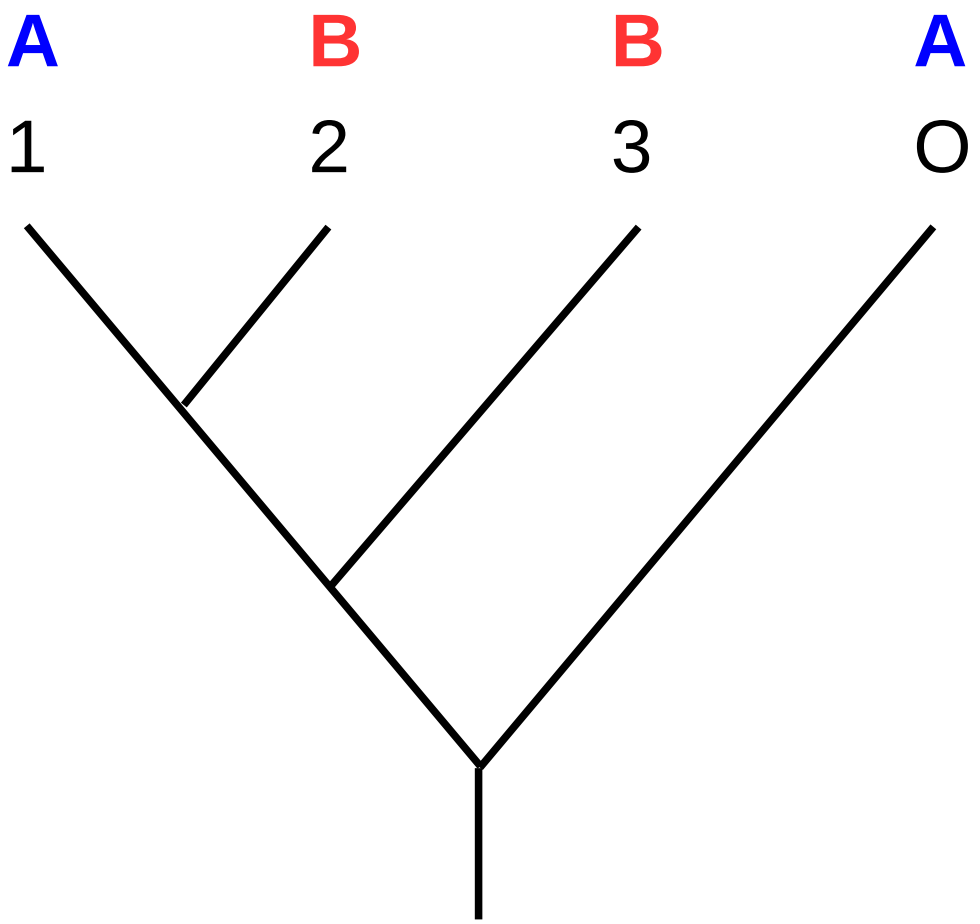
# BUT gene tree may not =  species tree!



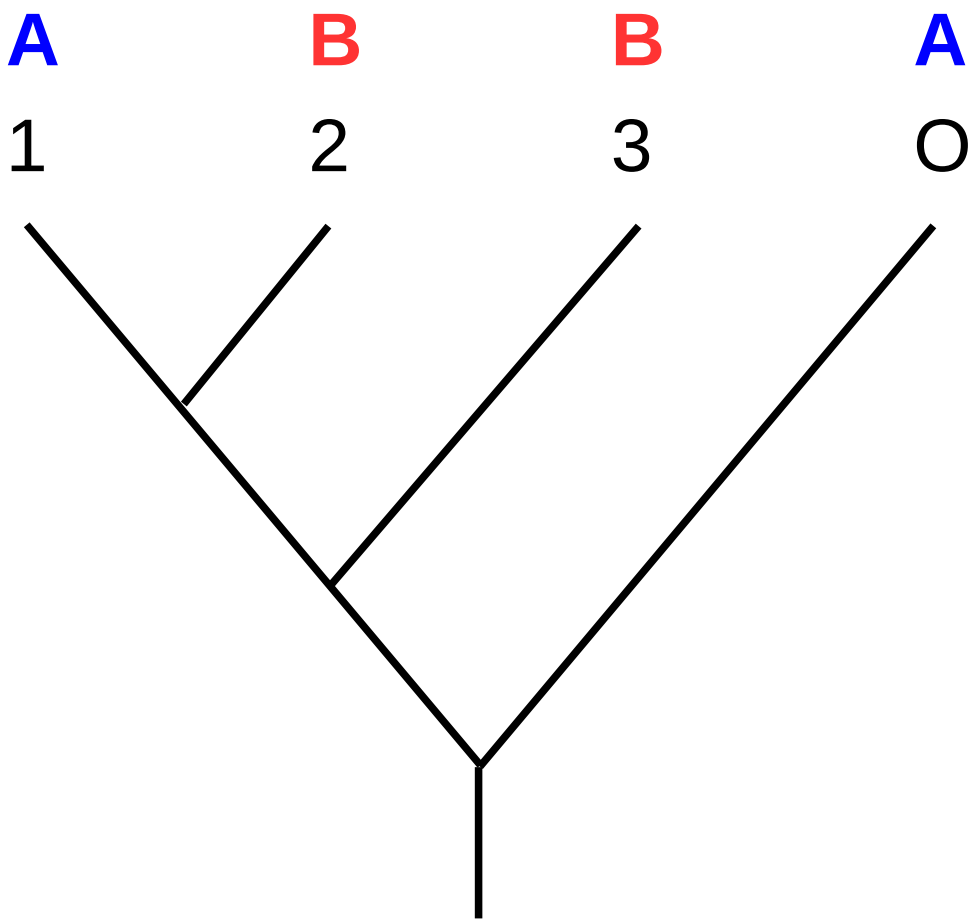
# BUT gene tree may not = species tree!



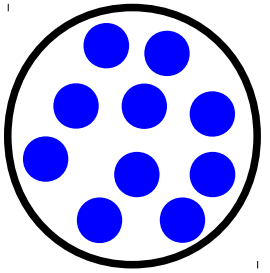
# Incomplete lineage sorting



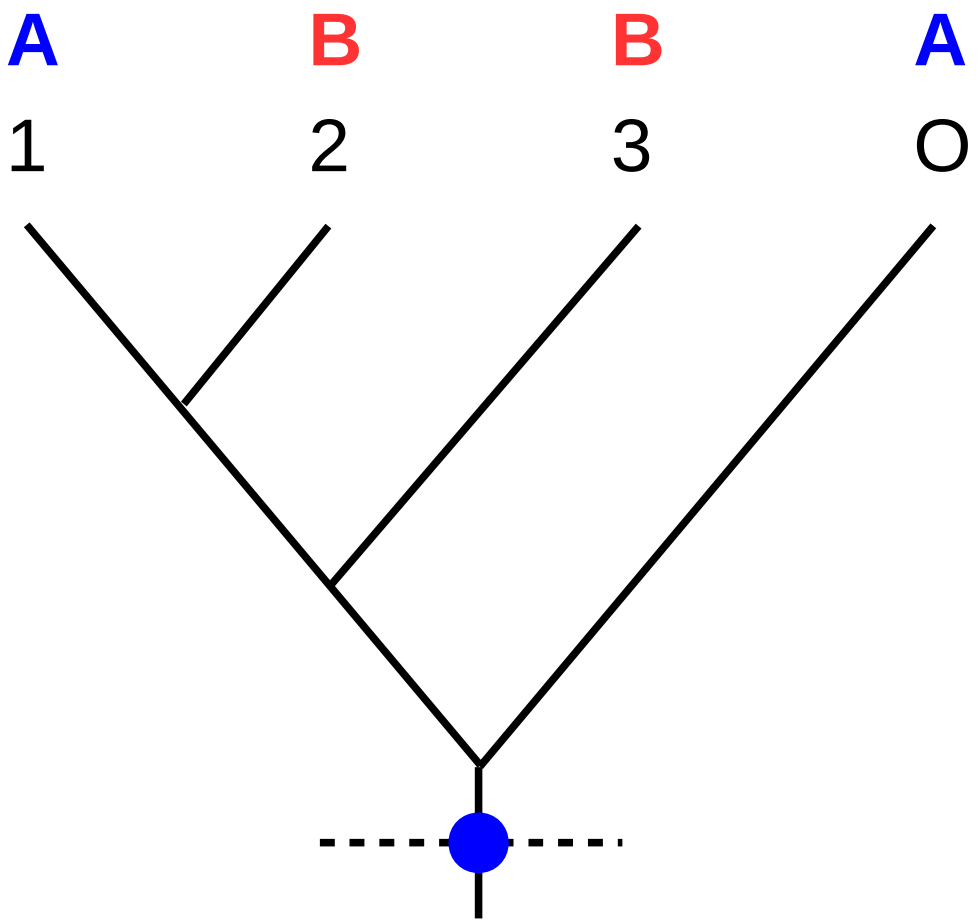
# Incomplete lineage sorting

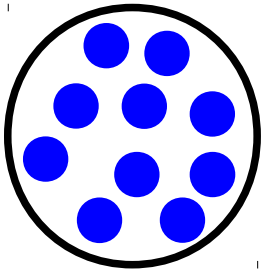



**(retention of ancestral polymorphism)**

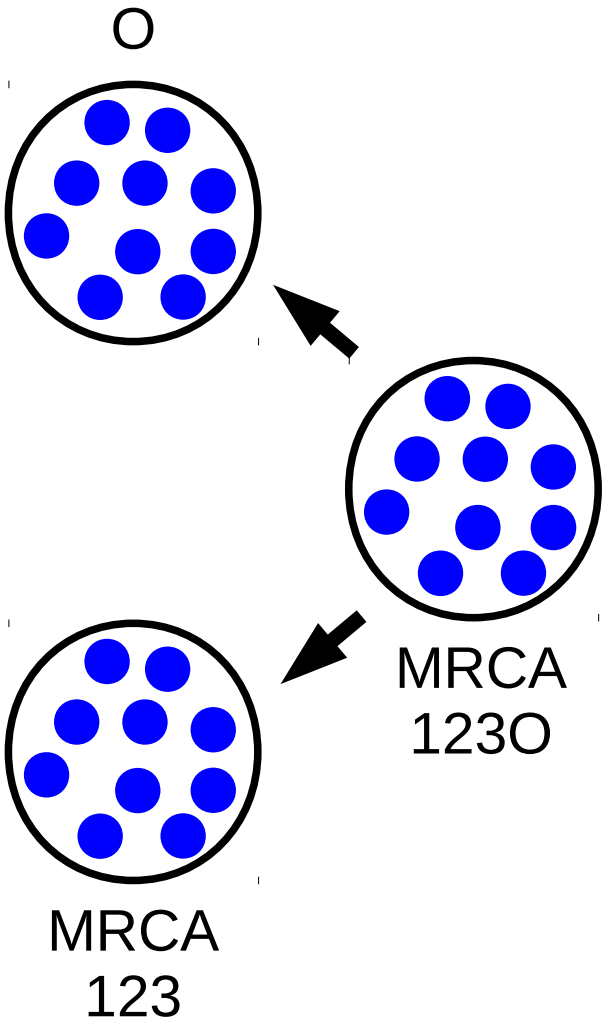


MRCA
123O

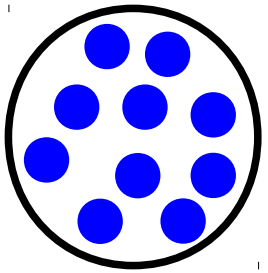

A
B
B
A
1
2
3
O

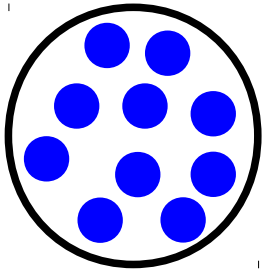


MRCA
123O

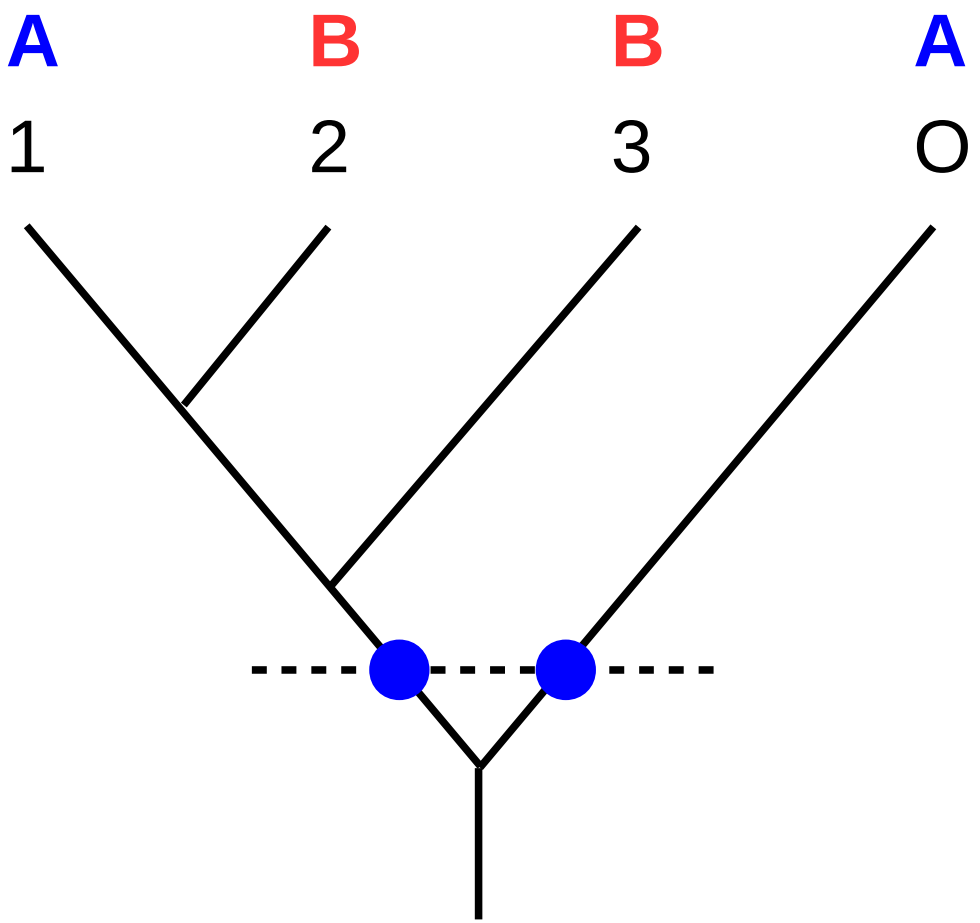

O
MRCA
123O
MRCA
123

O

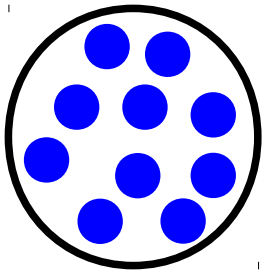


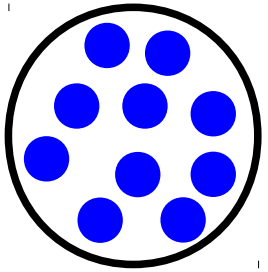


MRCA
123


A
B
B
A
1
2
3
O

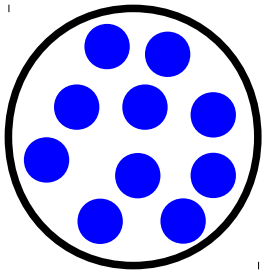
O

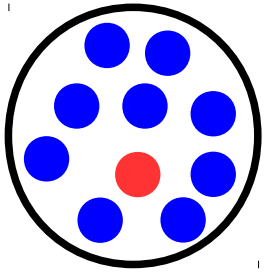




MRCA
123

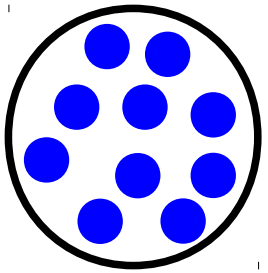
O

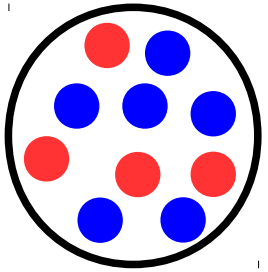




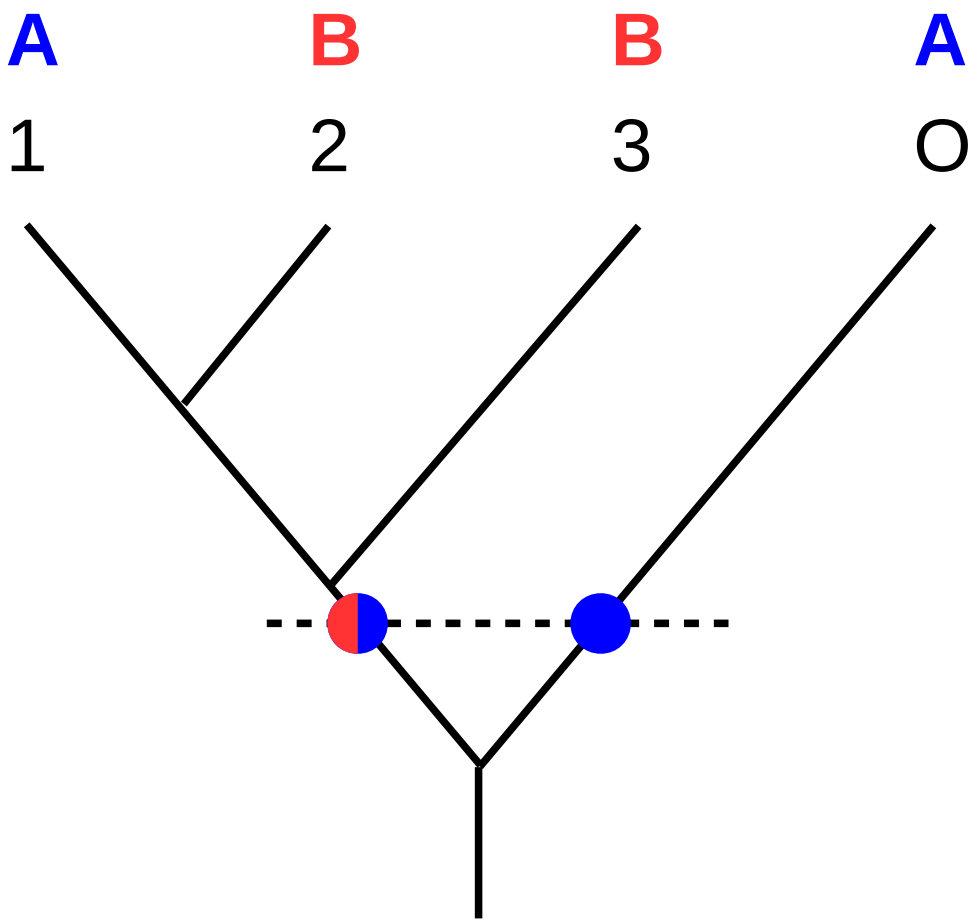
MRCA
123

O

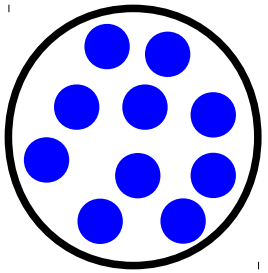


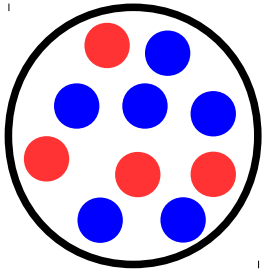


MRCA
123

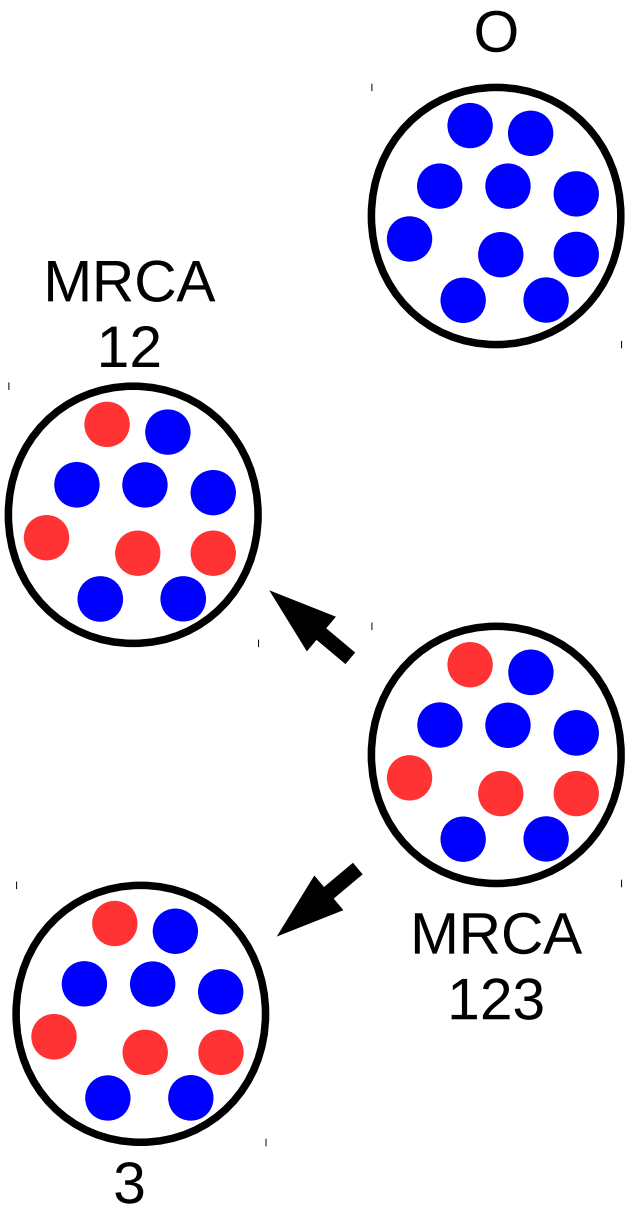

A
B
B
A
1
2
3
O

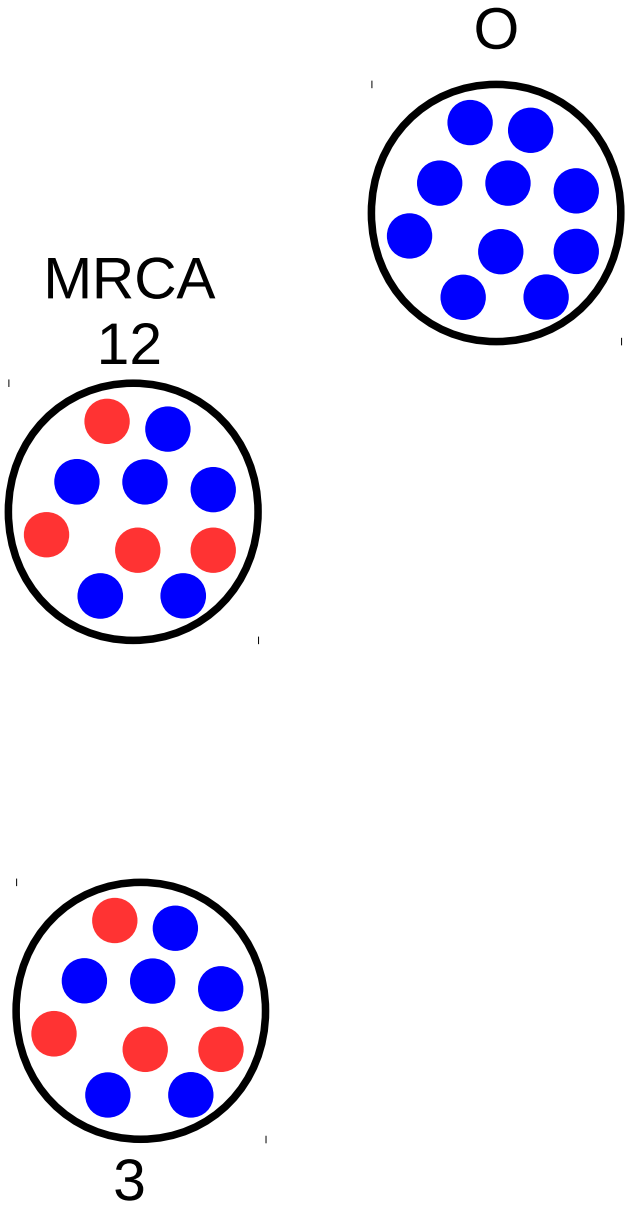
O

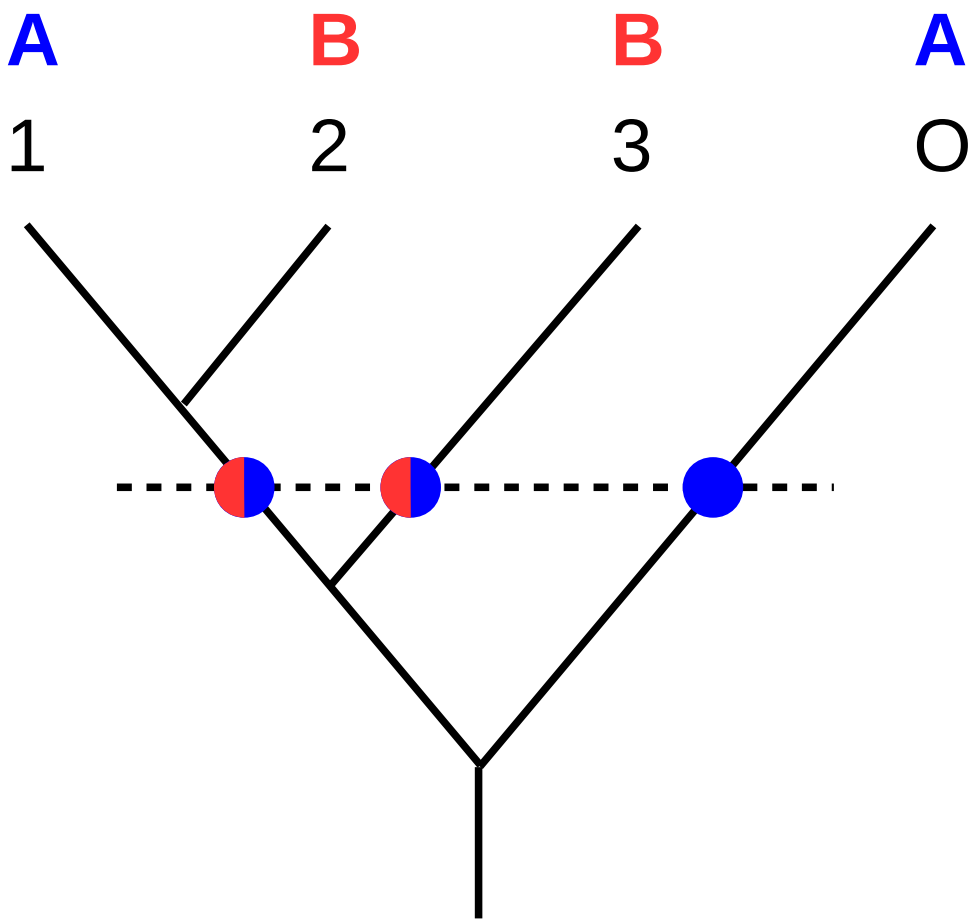


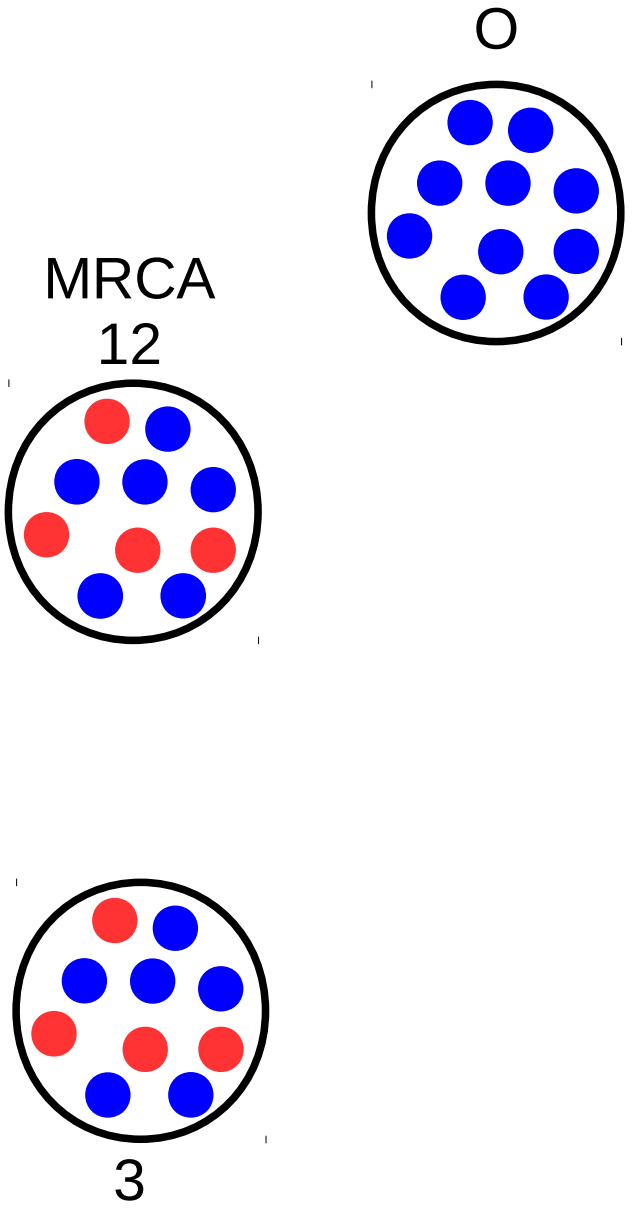


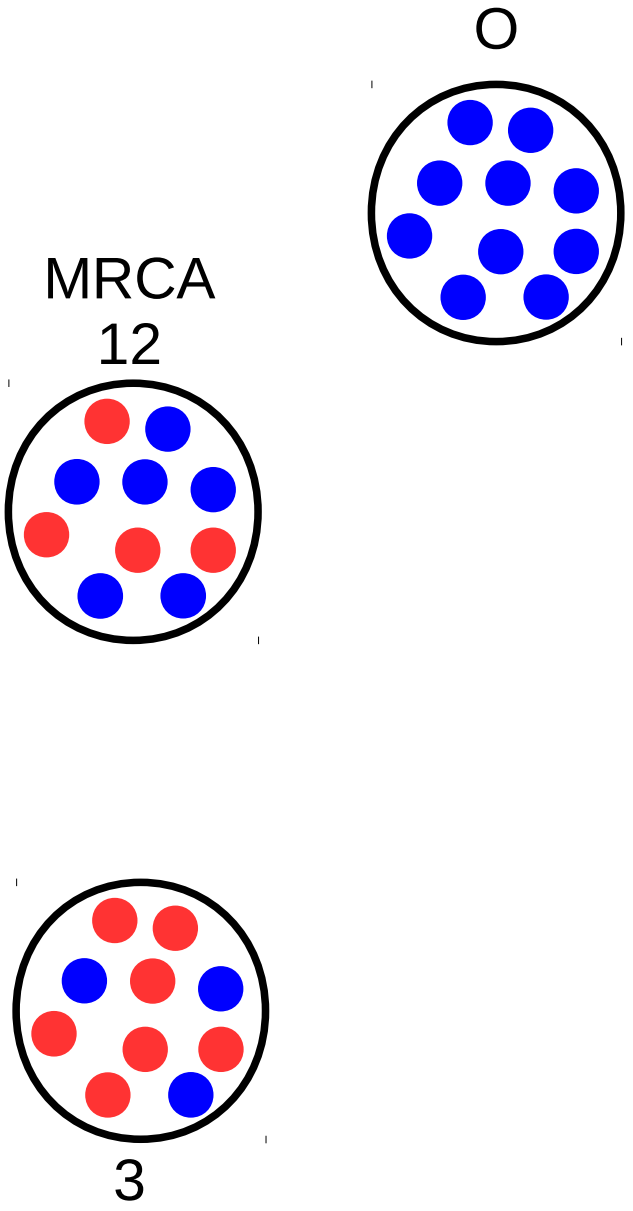
MRCA
123

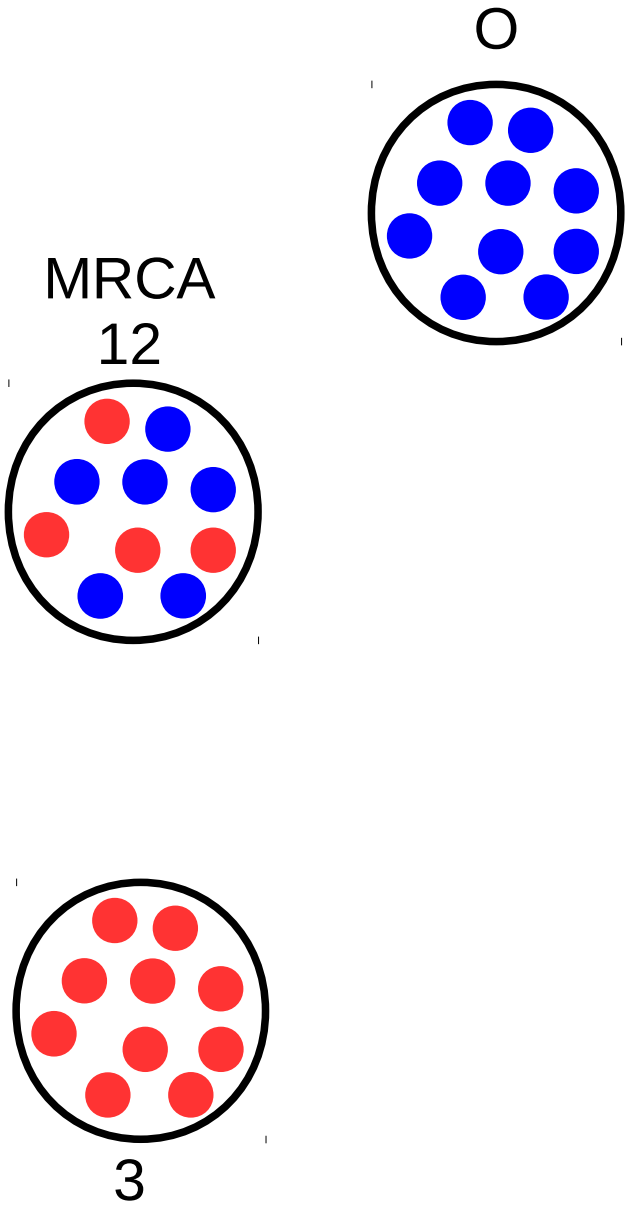

O
MRCA
12
MRCA
123
3

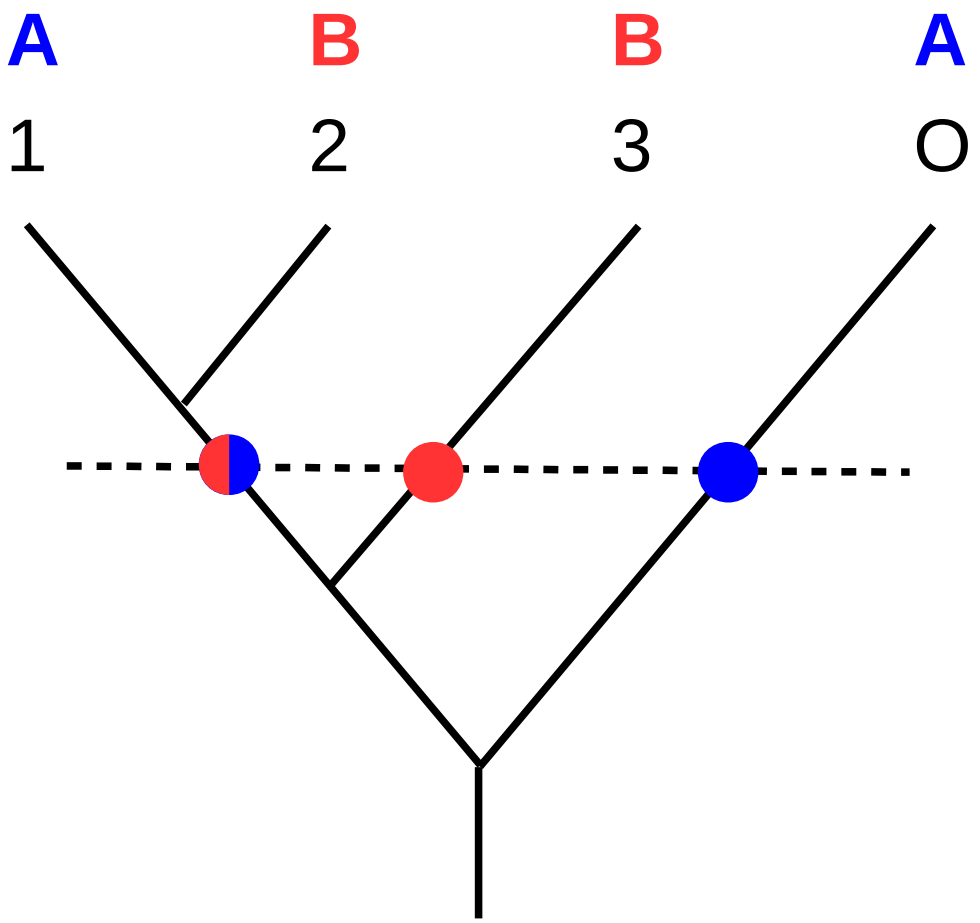

O
MRCA
12
3

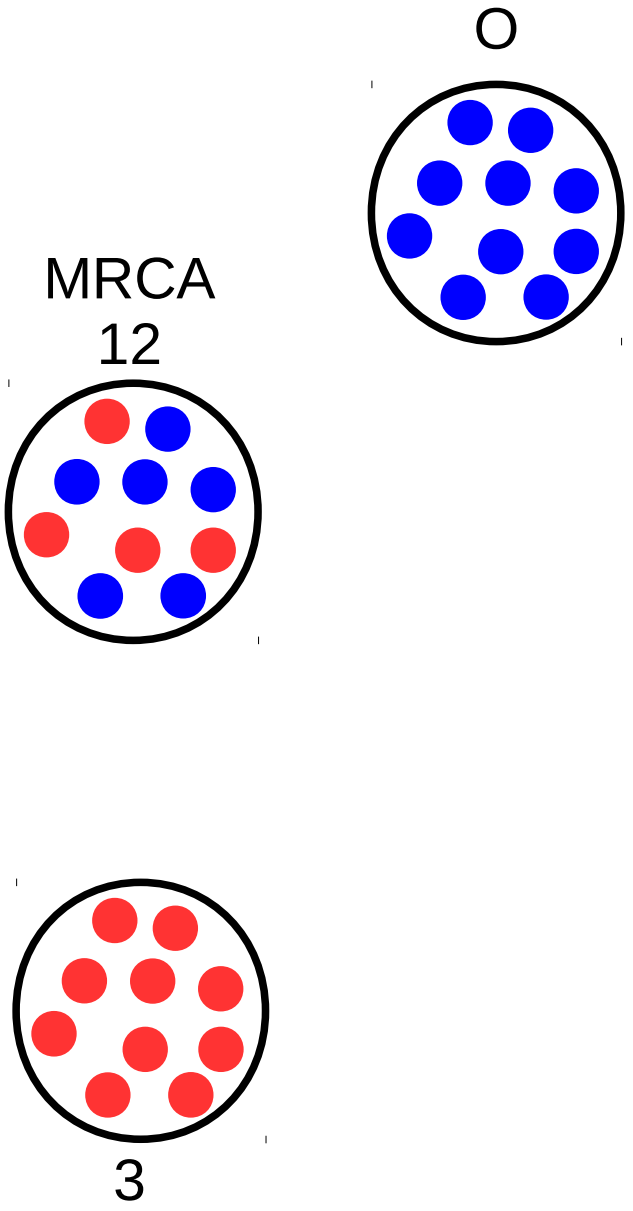

A
B
B
A
1
2
3
O

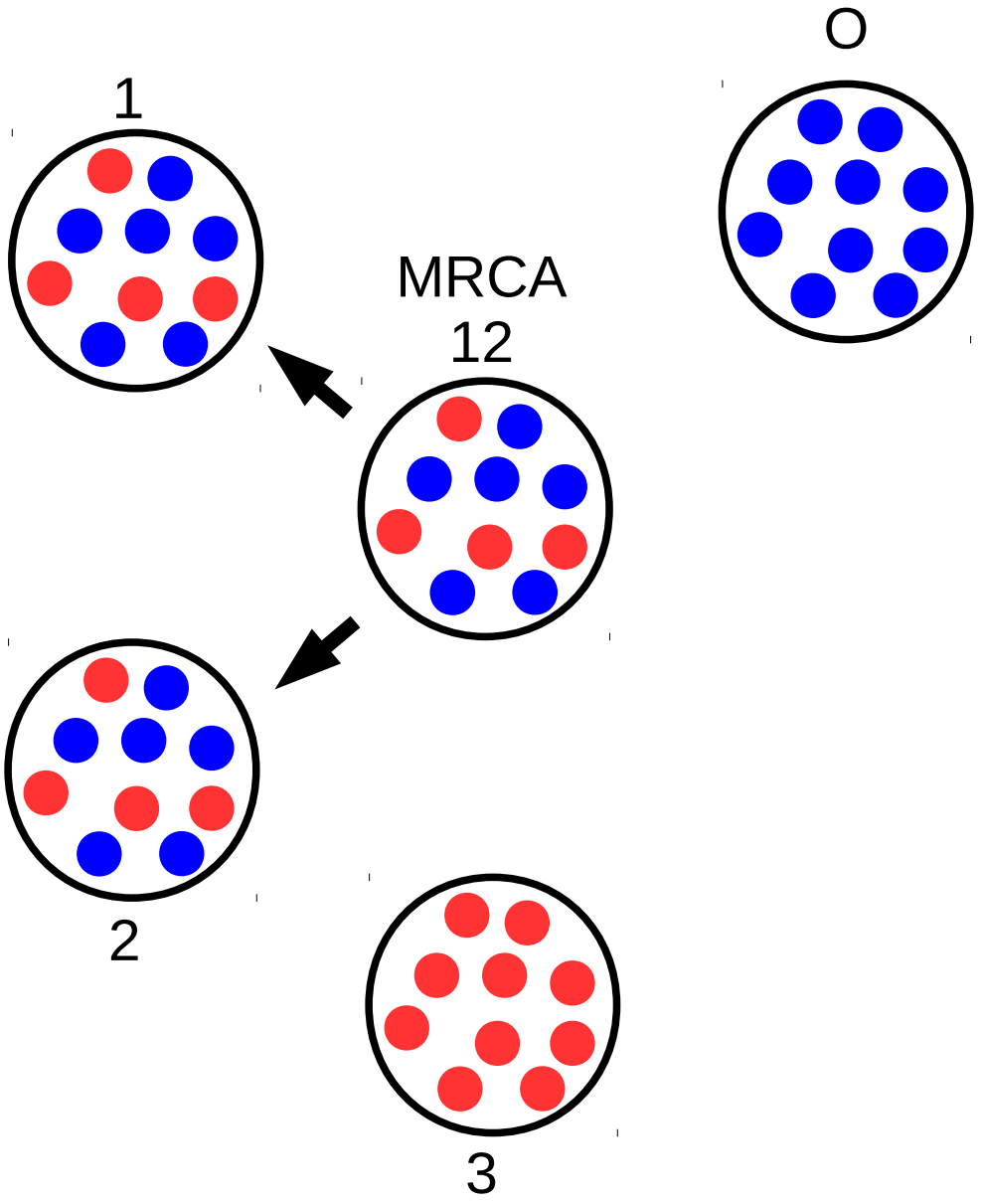

O
MRCA
12
3

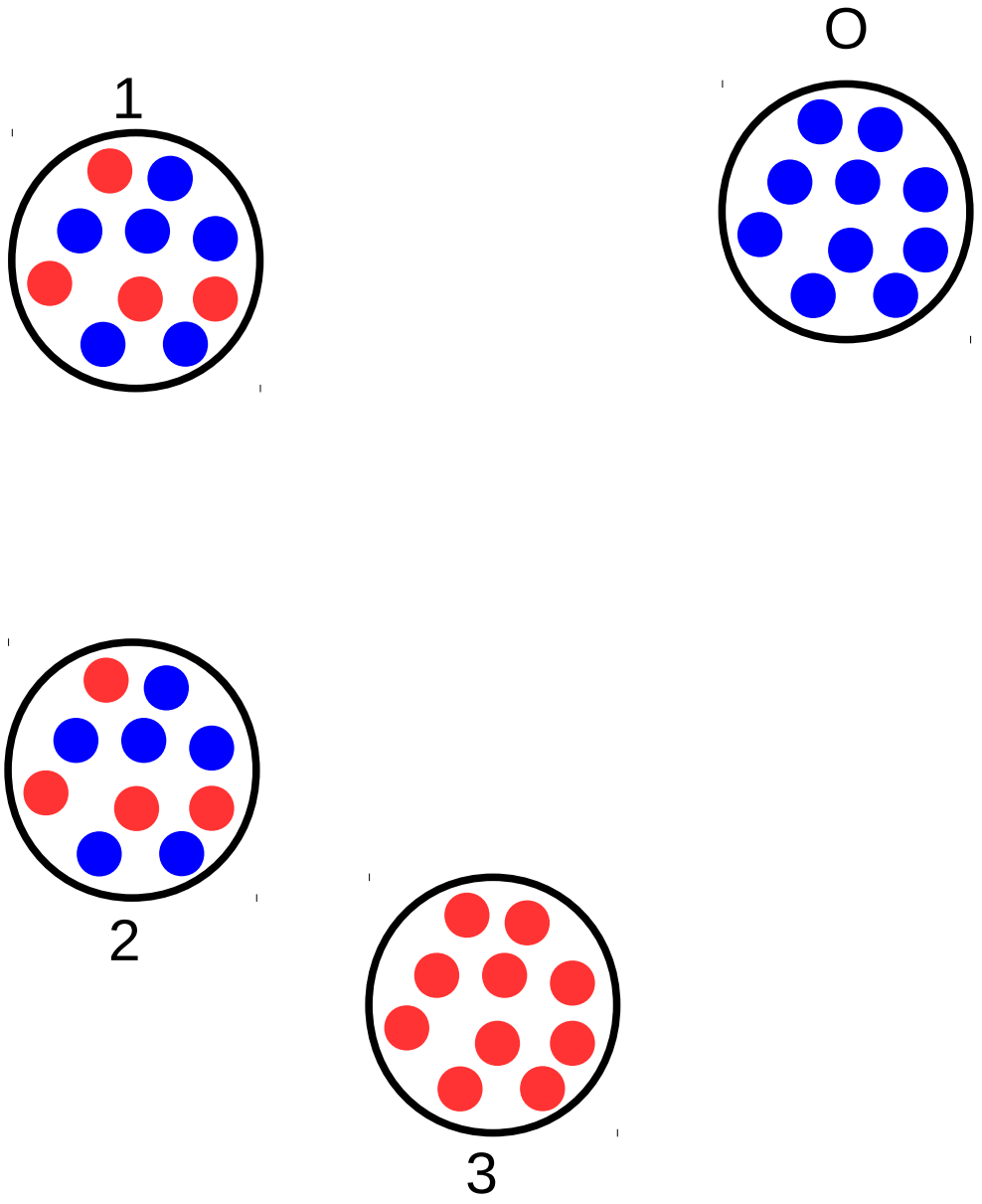

O
MRCA
12
3

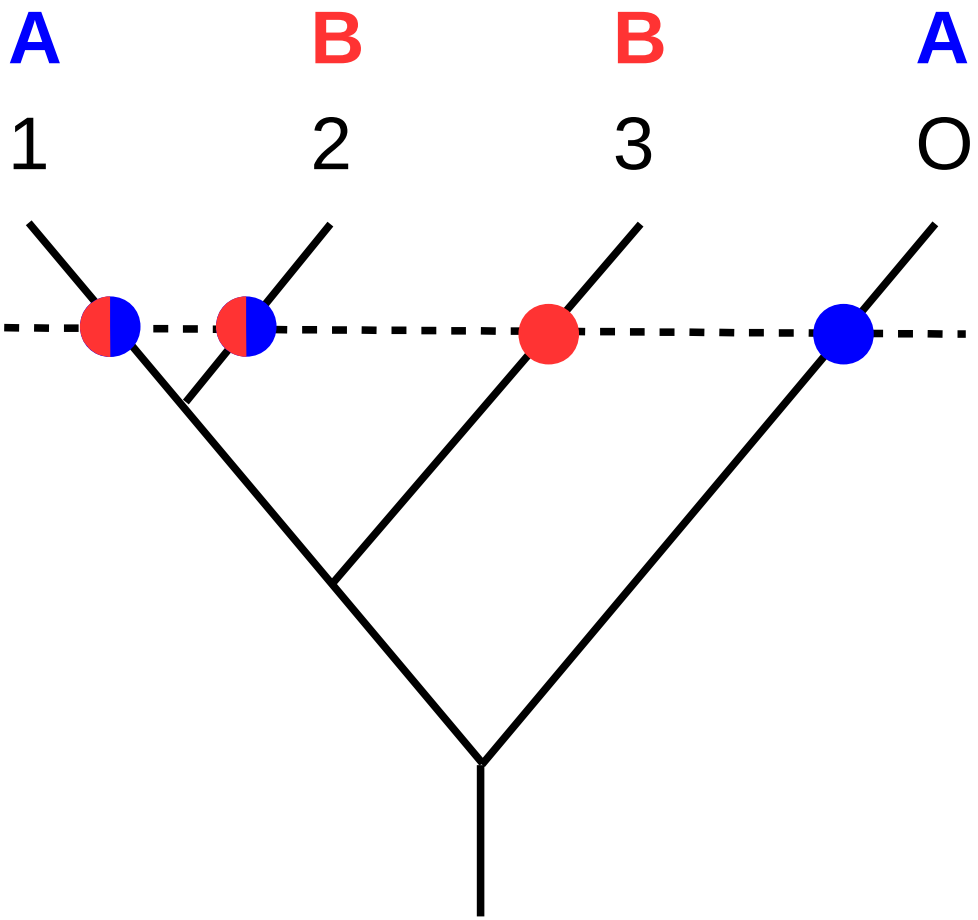

O
MRCA
12
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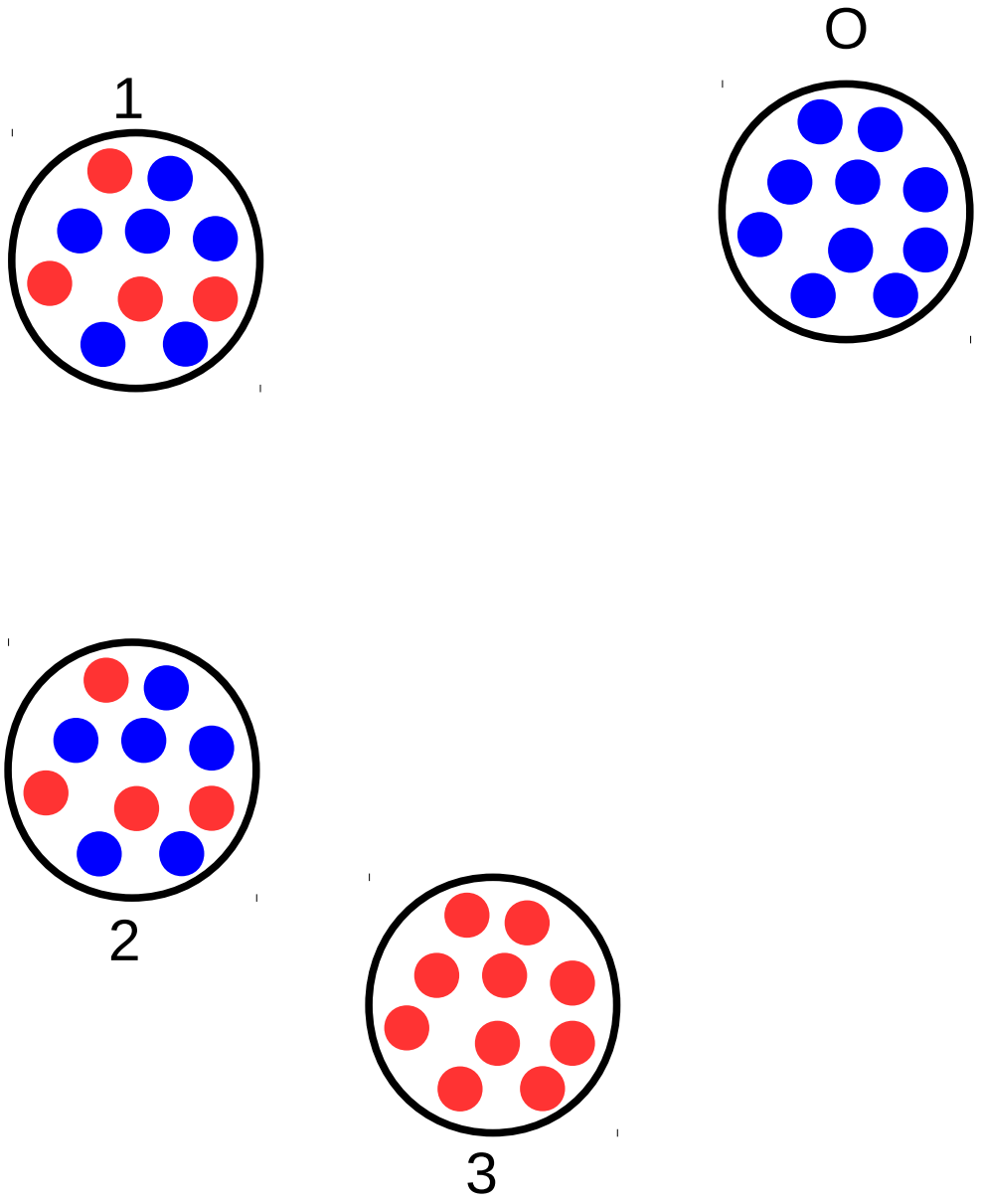

A
B
B
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1
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3
O

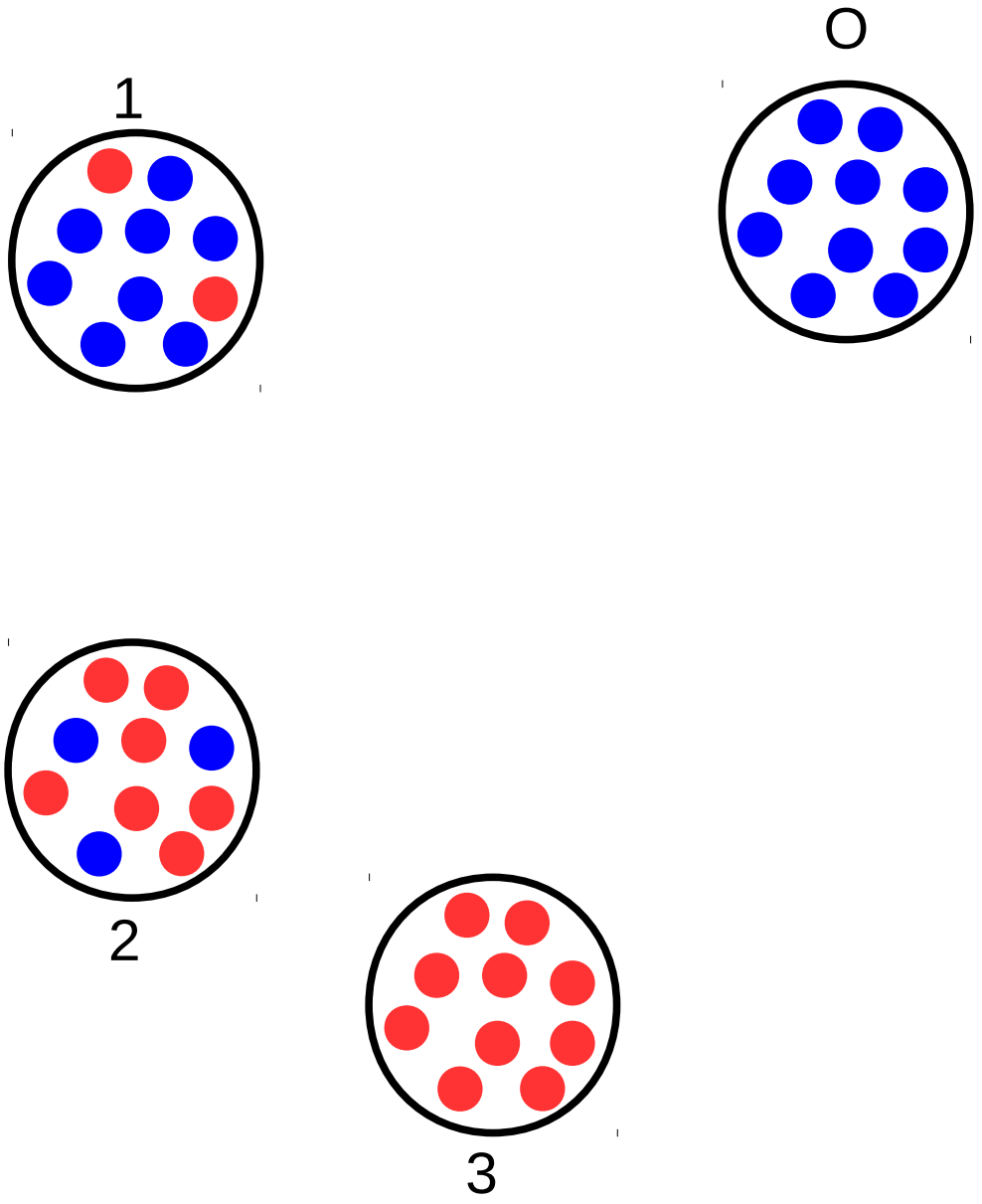

O
MRCA
12
3

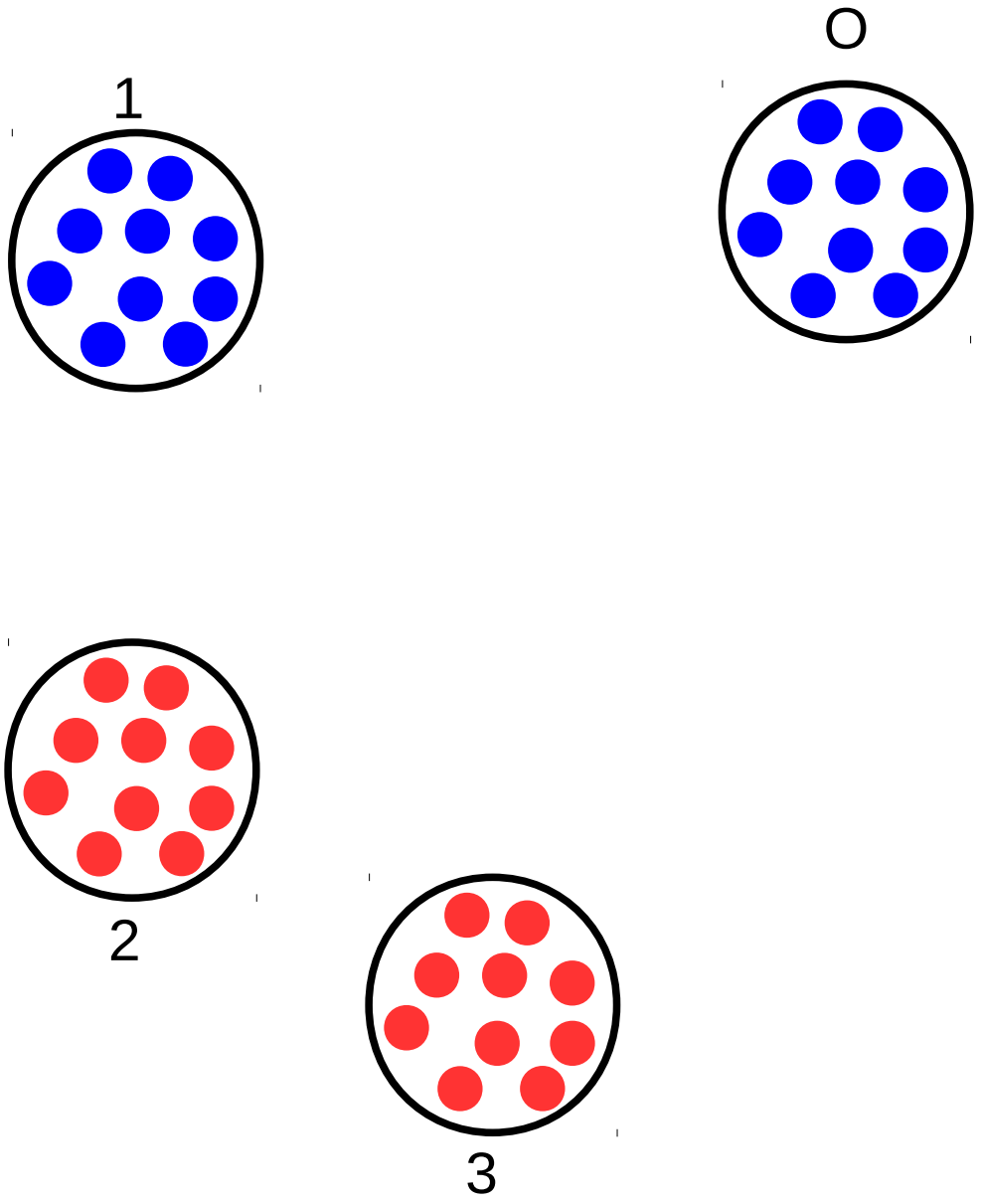

O
1
MRCA
12
2
3


O
1
2
3

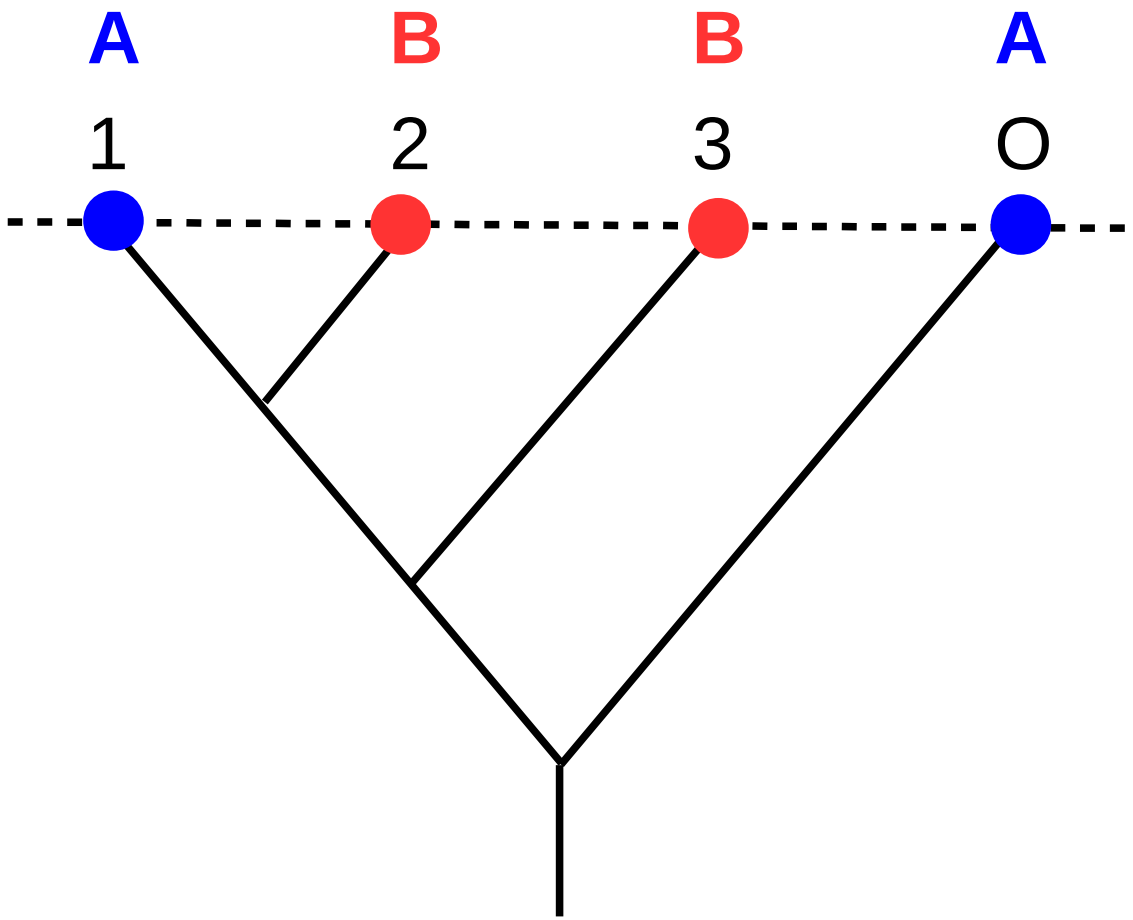

A
B
B
A
1
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3
O


O
1
2
3


1
O
2
3


1
O
2
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## Retention of ancestral polymorphism + drift



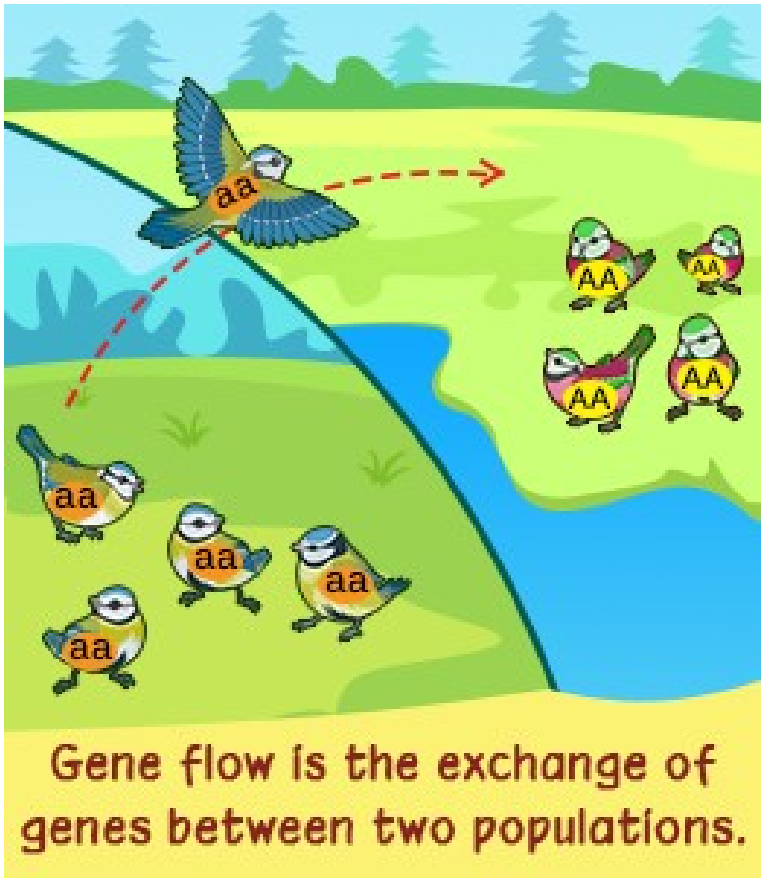

## Gene tree does not = species tree

# Incomplete lineage sorting

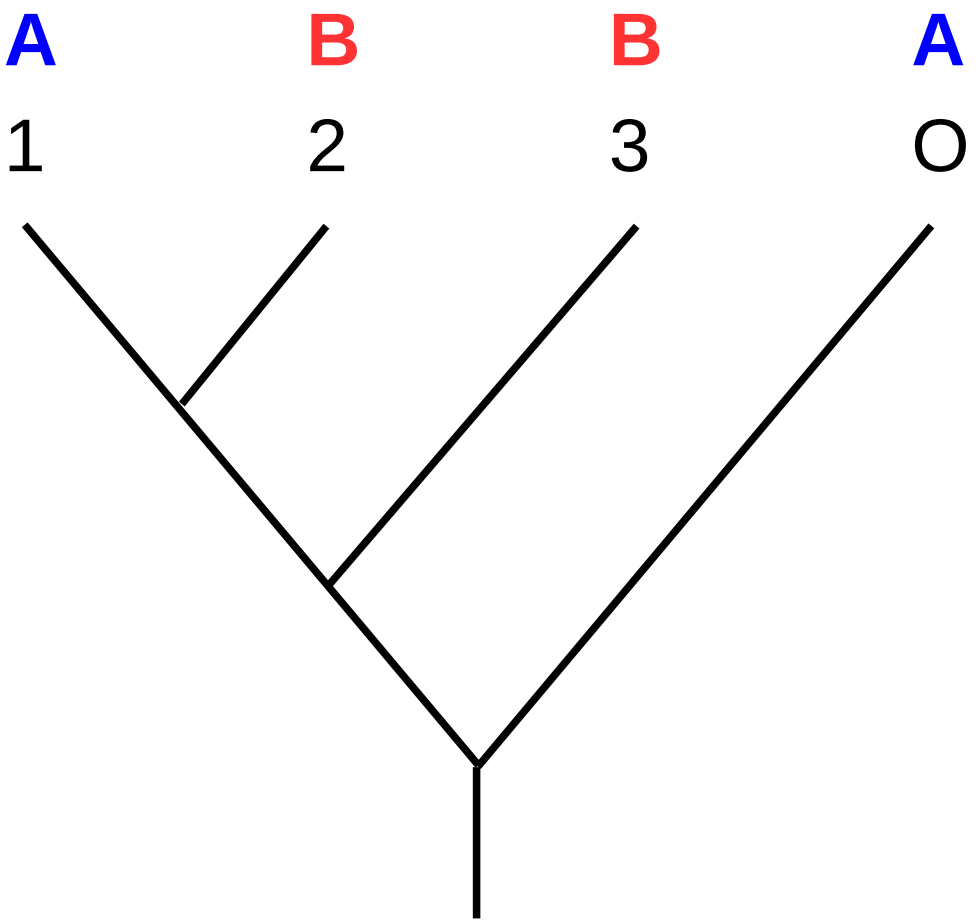
- Retention of ancestral polymorphism along successive nodes of the population/species tree

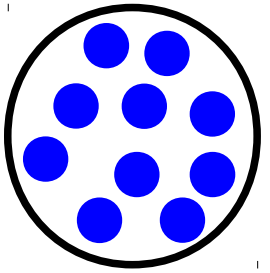
  Or...

- Failure to reach reciprocal monophyly before next speciation or population isolation event
- With or without random drift
- Result: gene trees may not = species tree

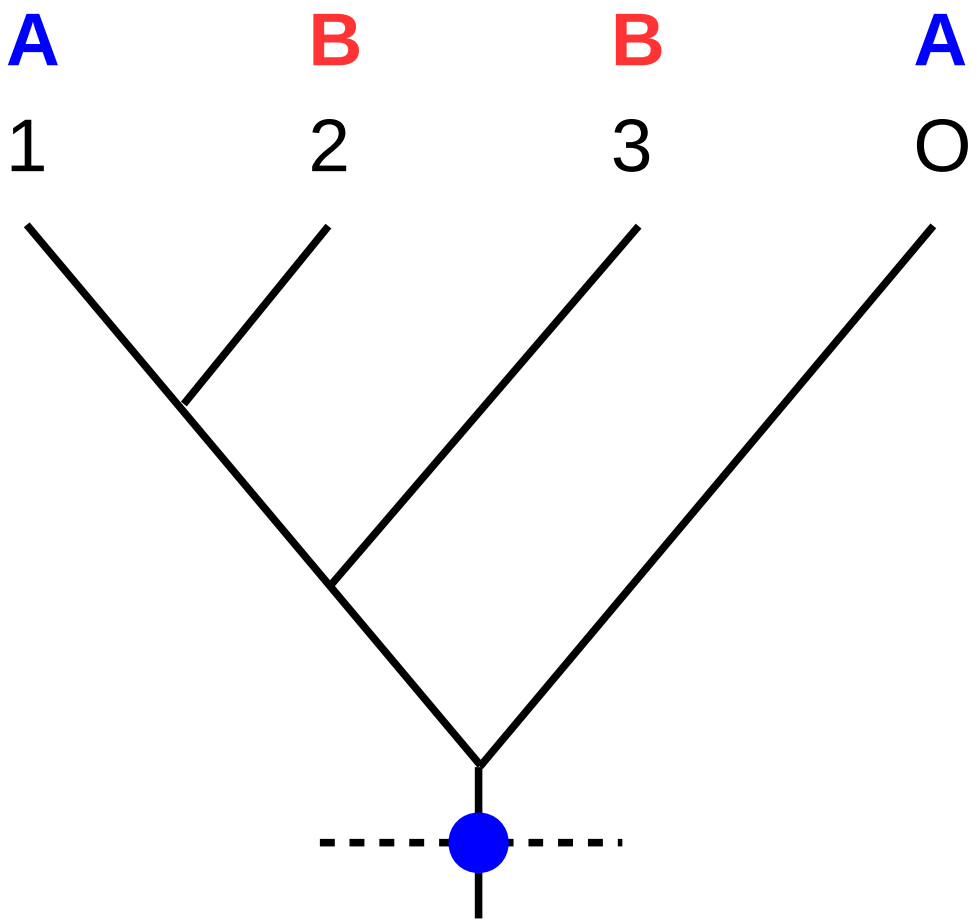

aa
AA
AA
AA
AA
aa
aa
aa
aa
Gene flow is the exchange of genes between two populations.

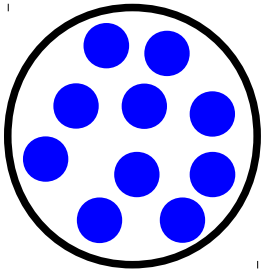
# Gene flow



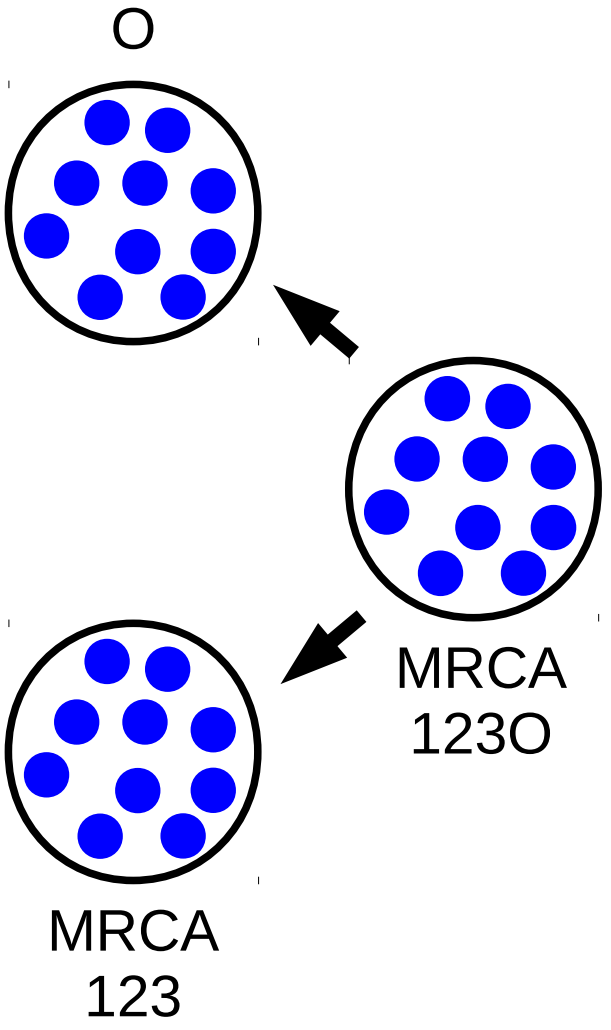


MRCA
123O

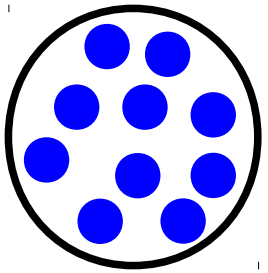

A
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B
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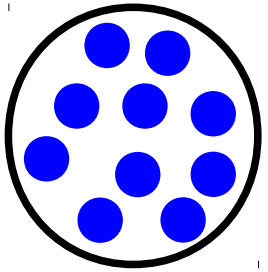


MRCA
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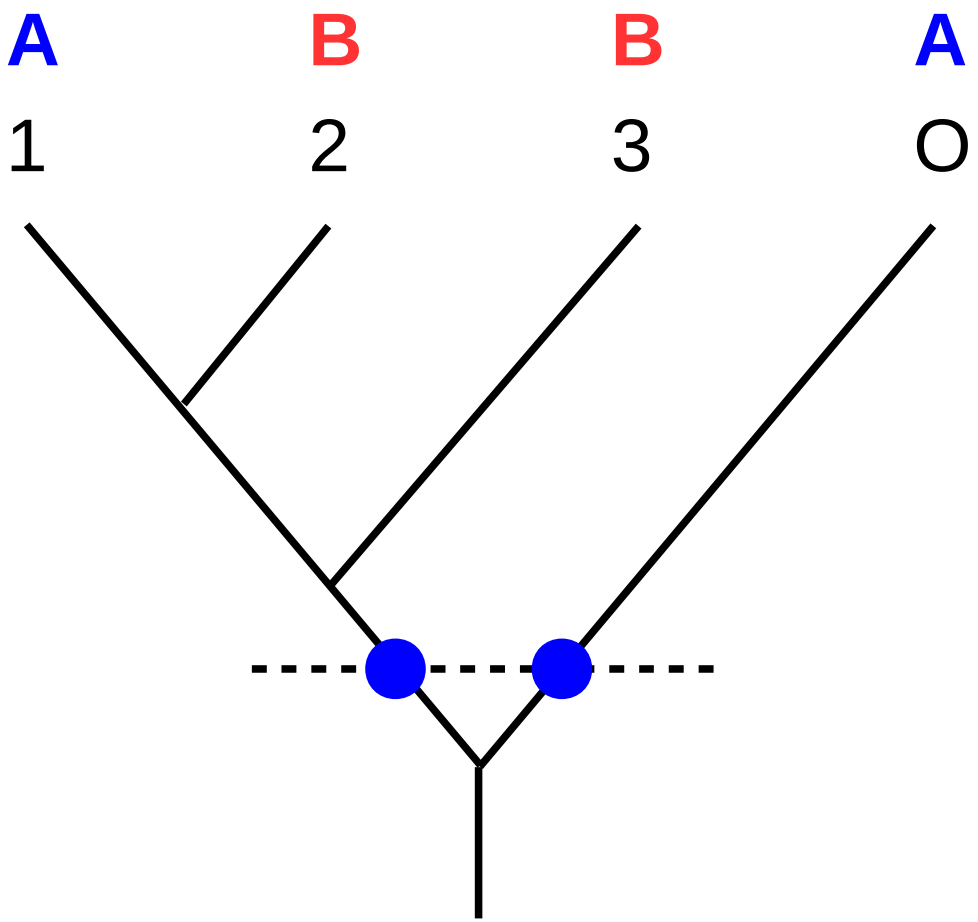

O
MRCA
123O
MRCA
123

O

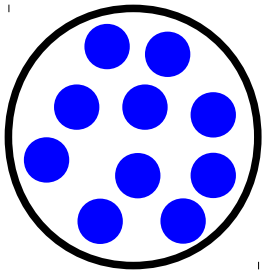


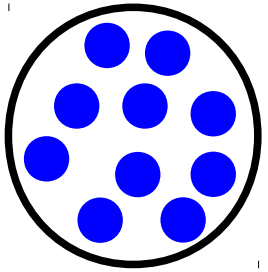


MRCA
123

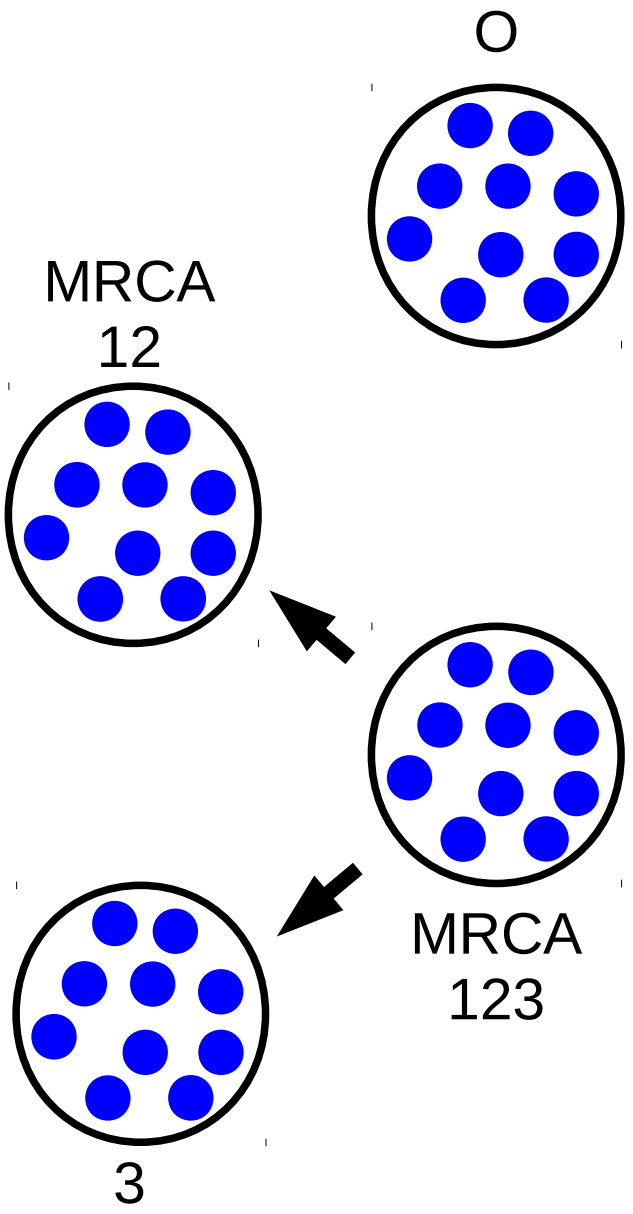

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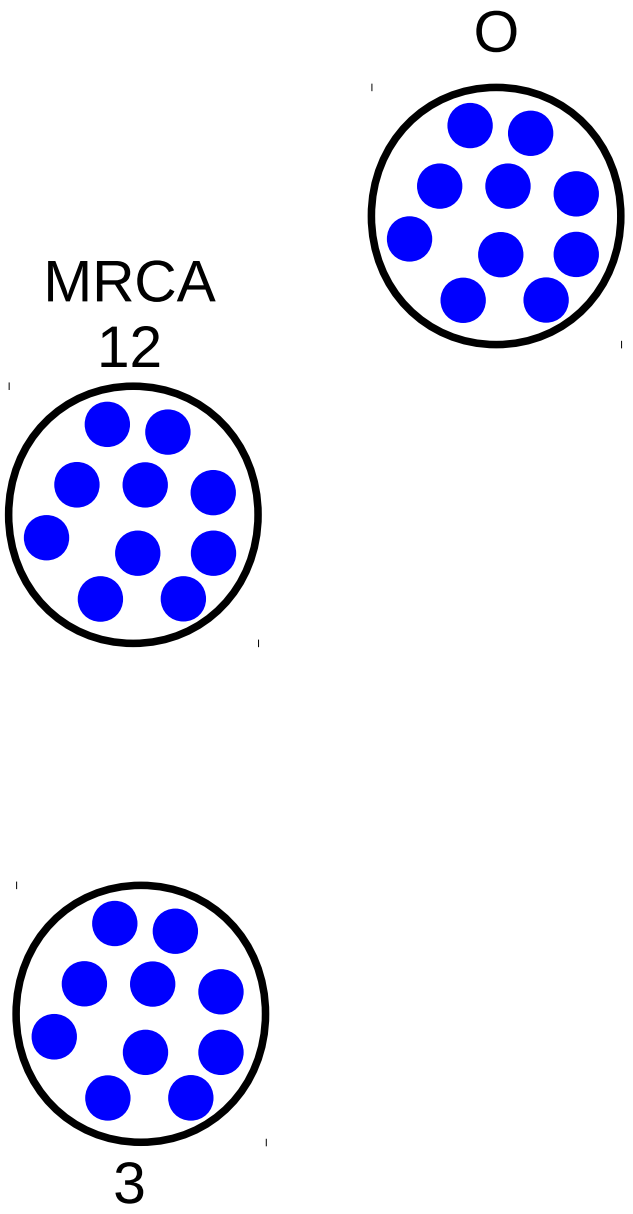
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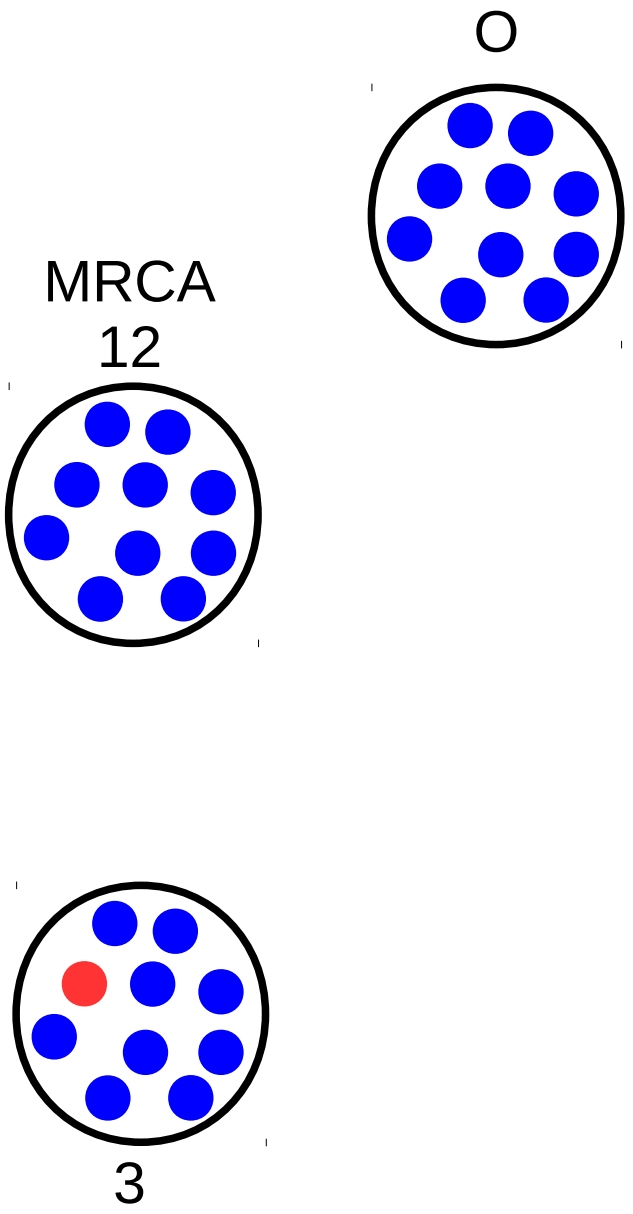


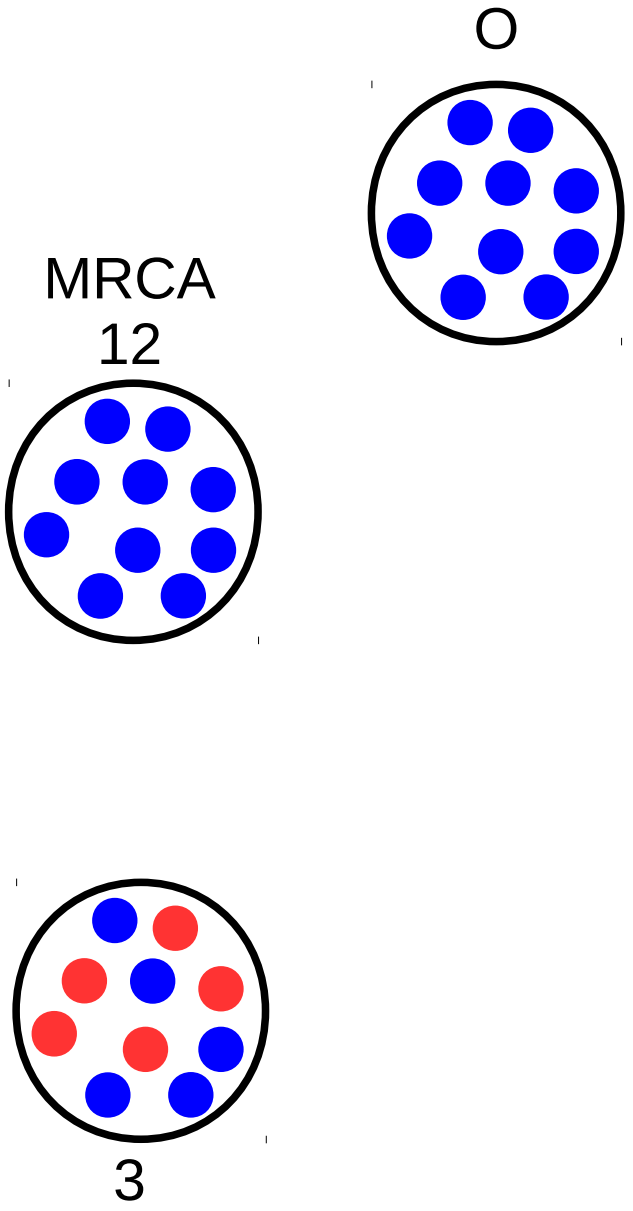


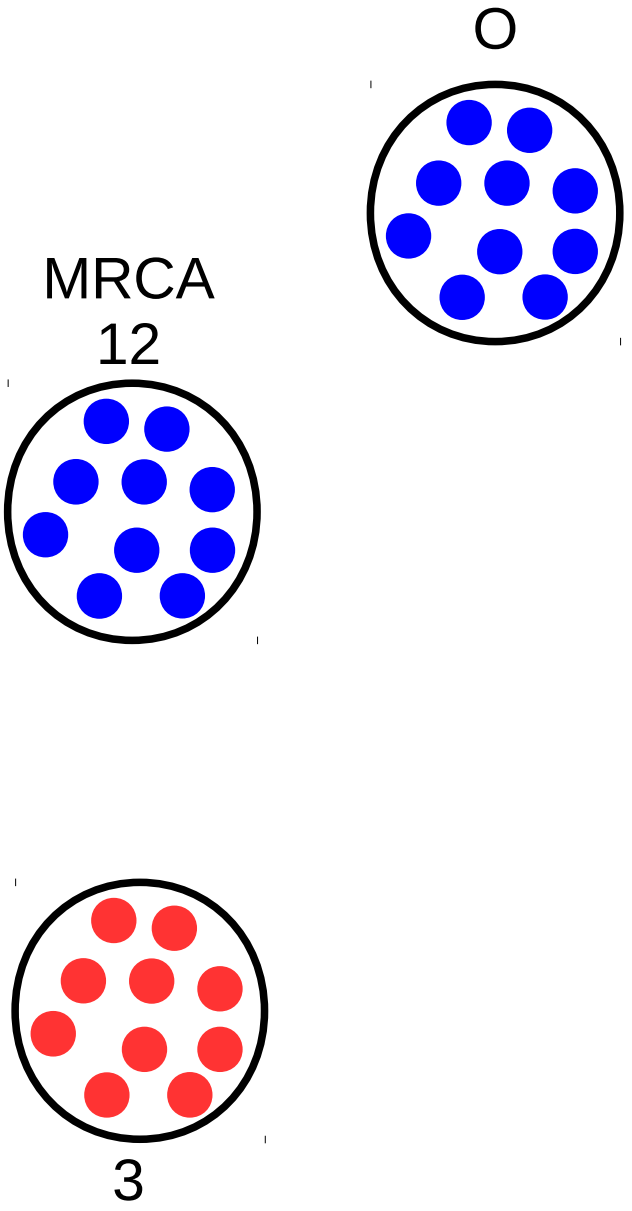
MRCA
123

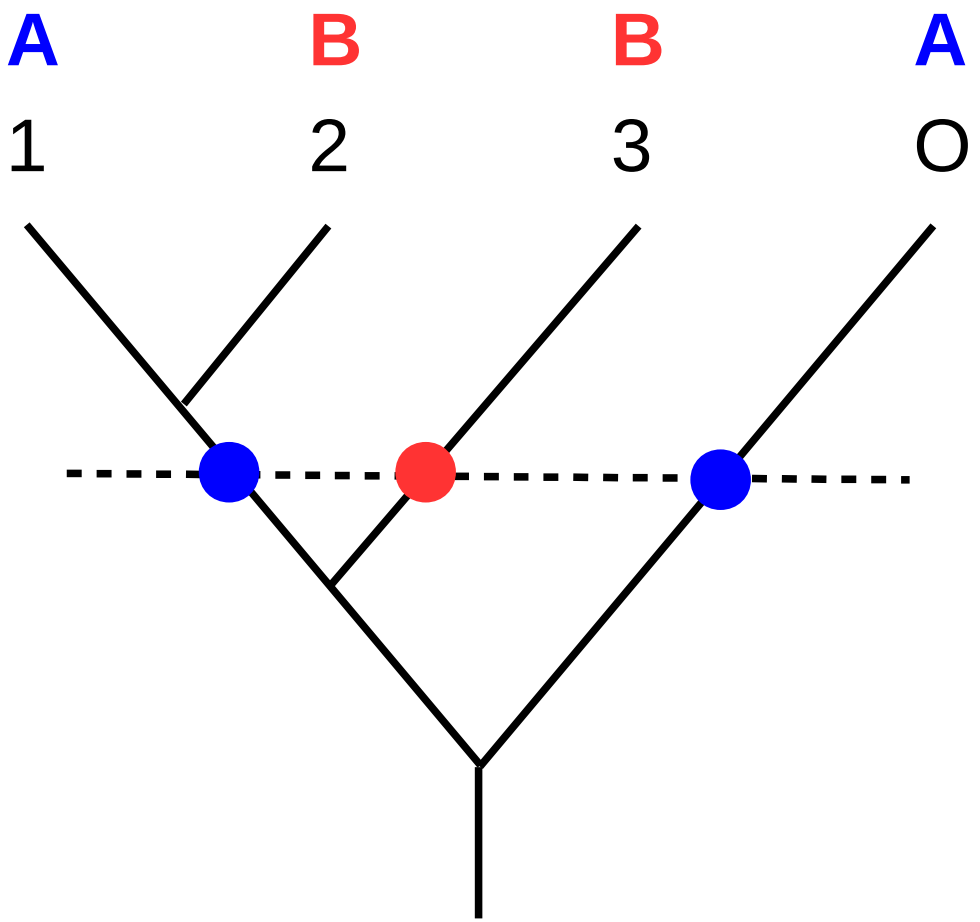

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MRCA
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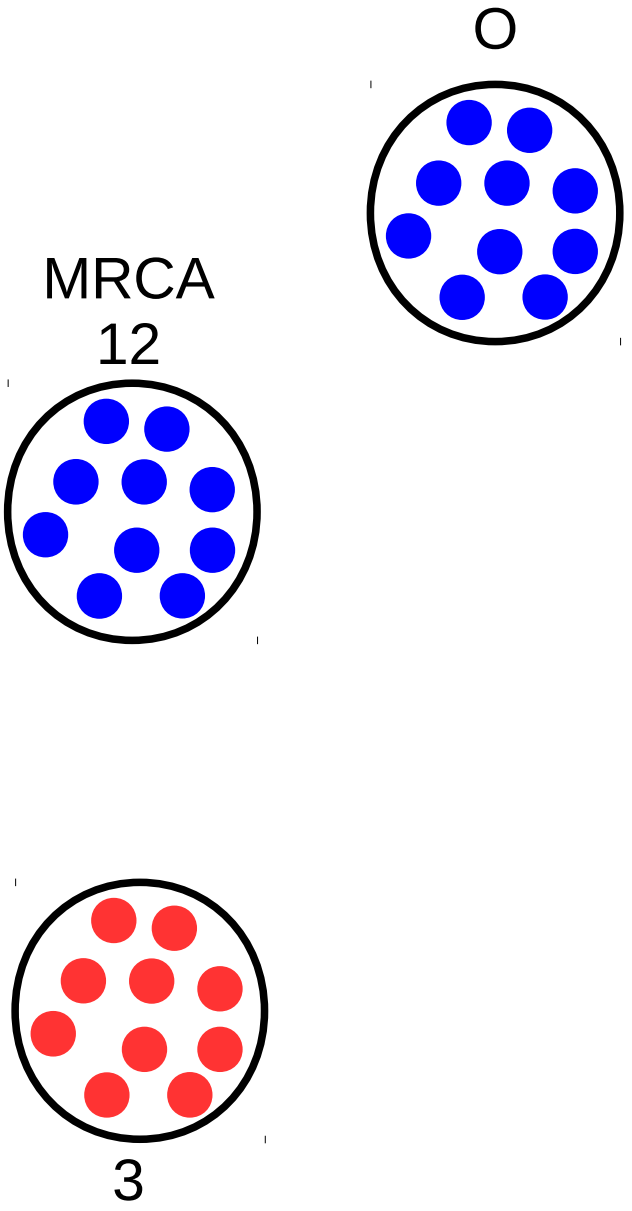

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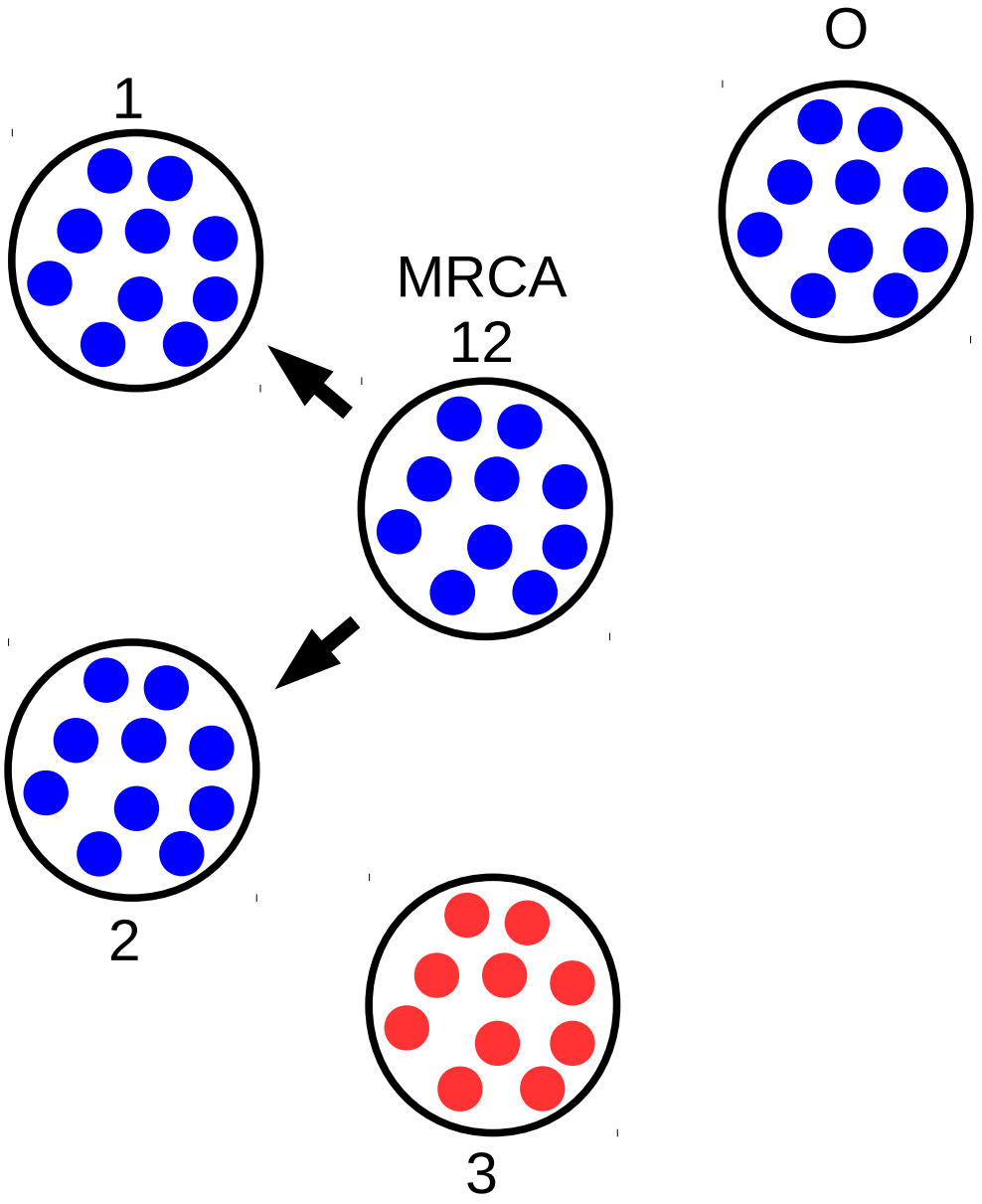

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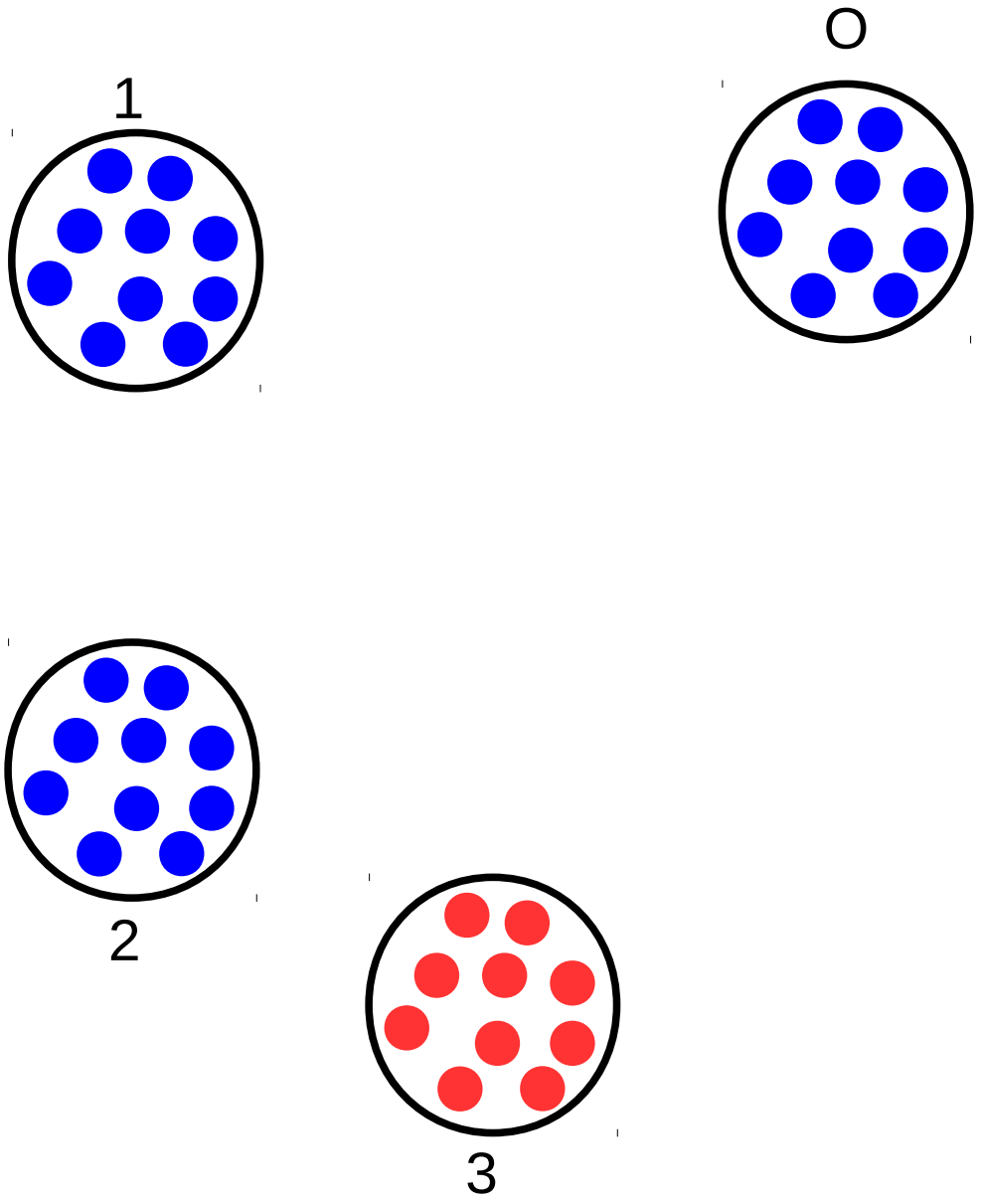

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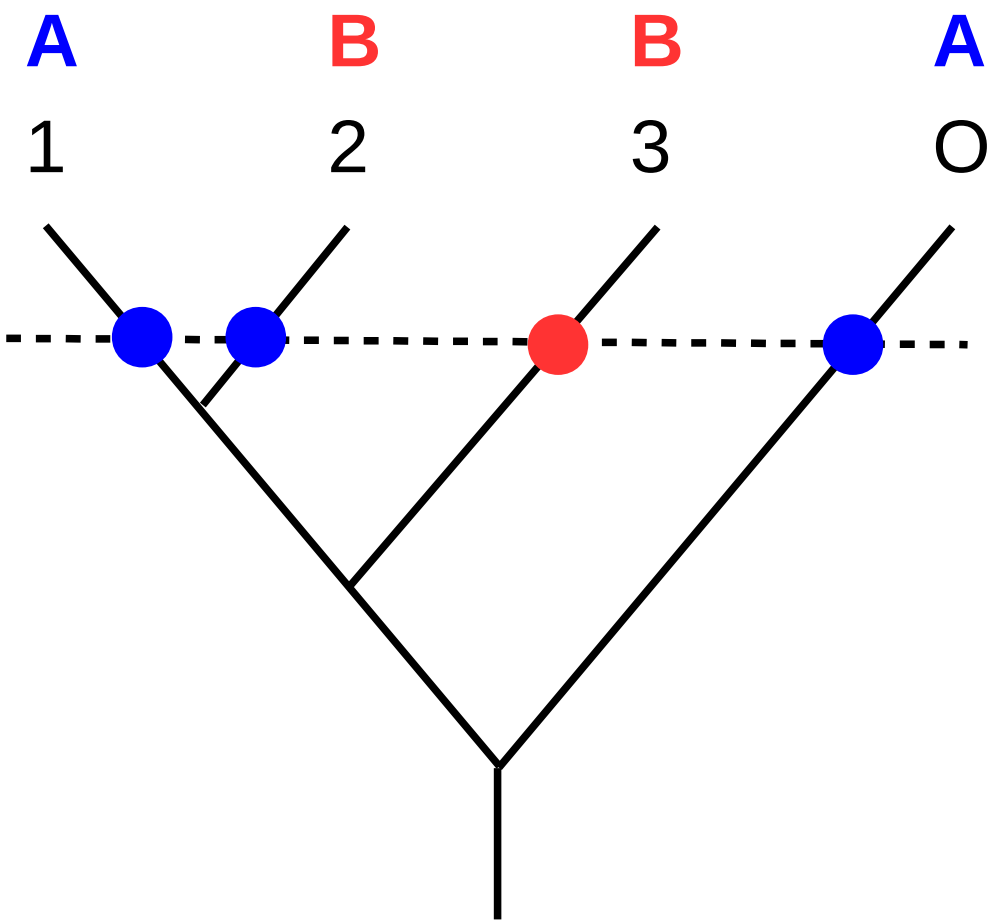

O
MRCA
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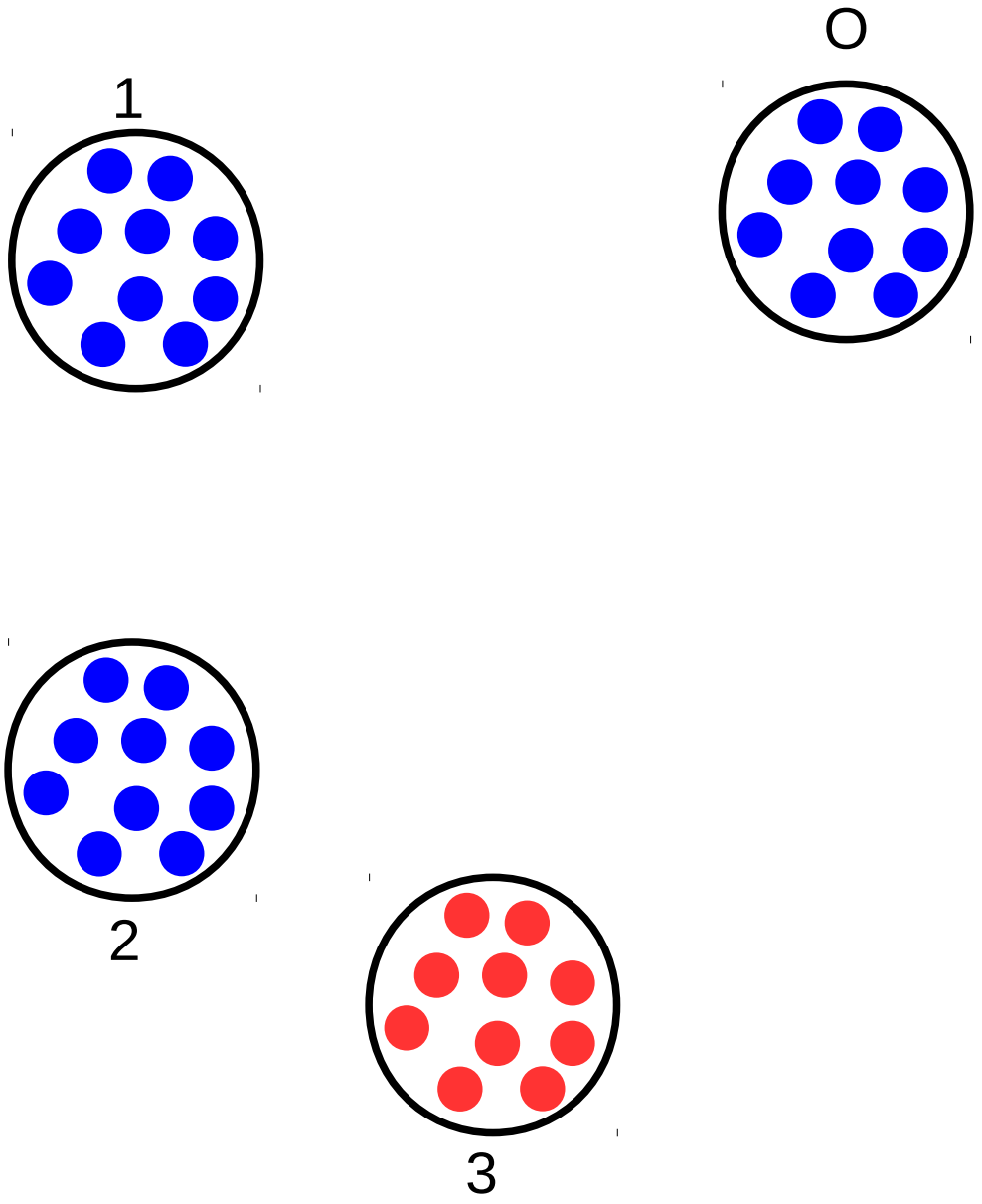

A
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B
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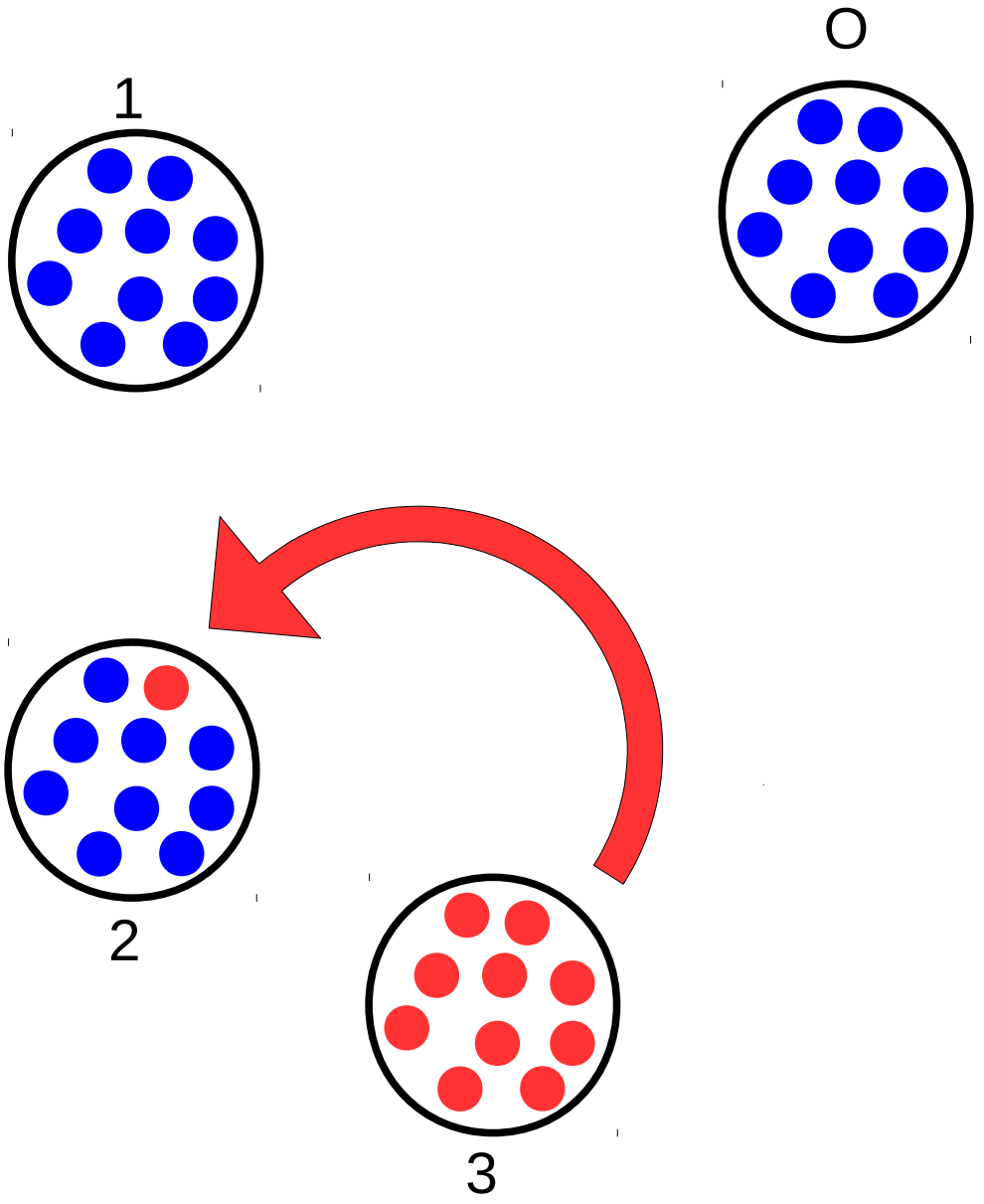

O
MRCA
12
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O
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MRCA
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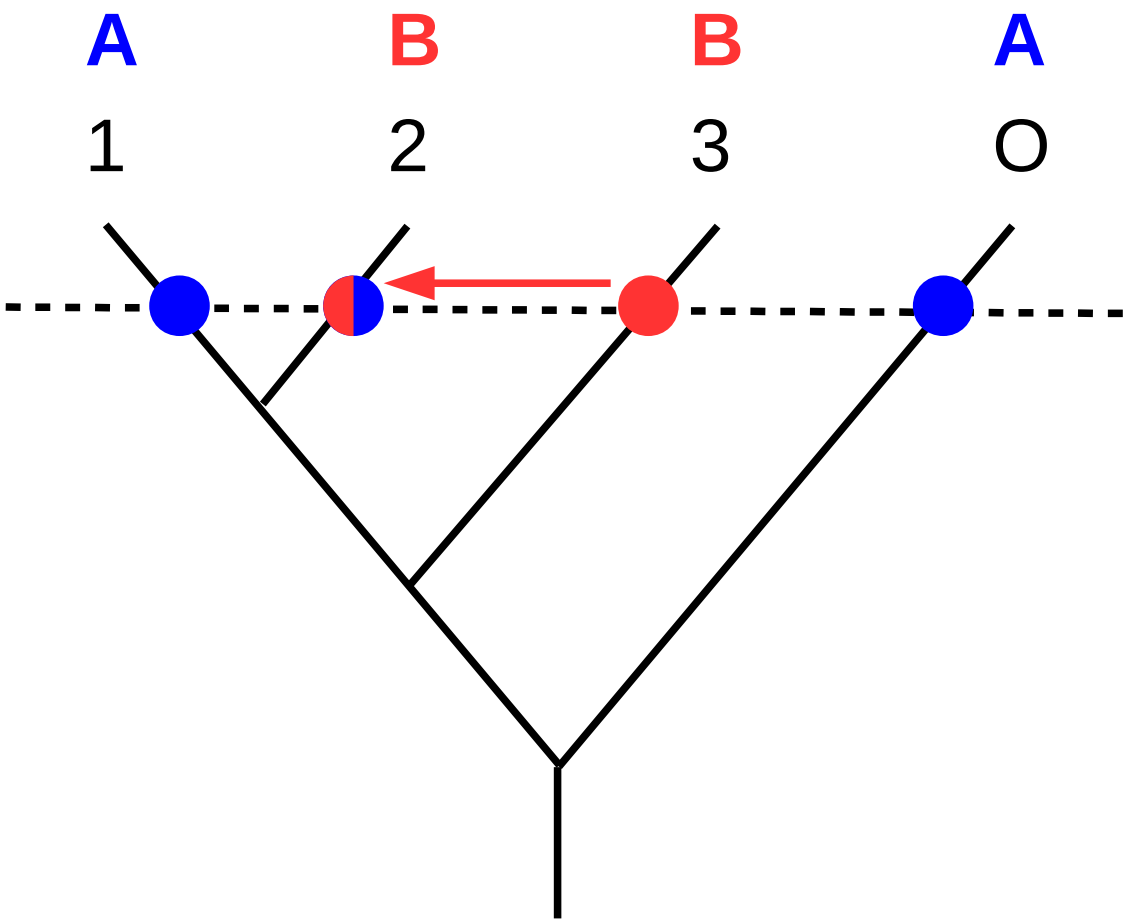

O
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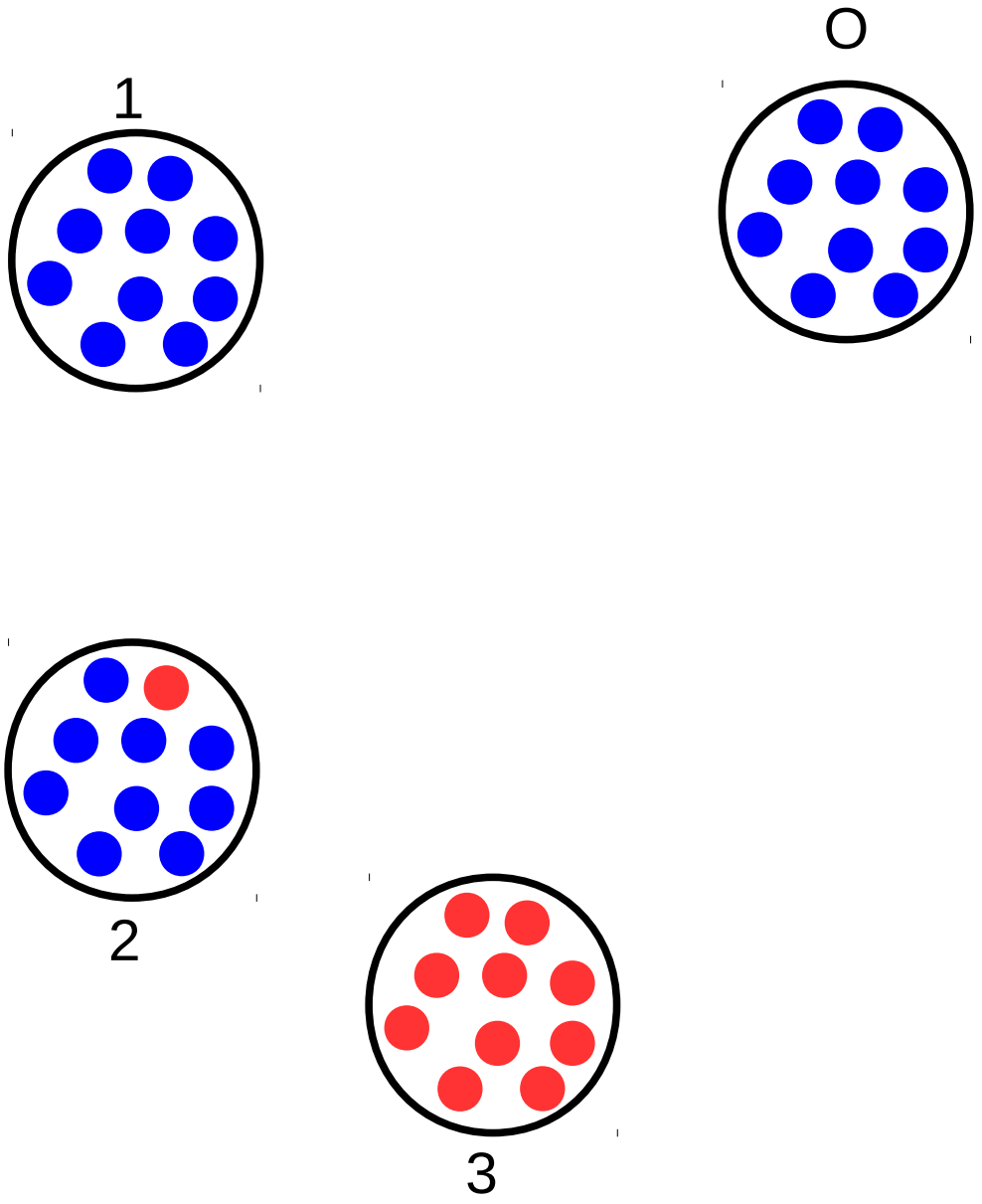

A
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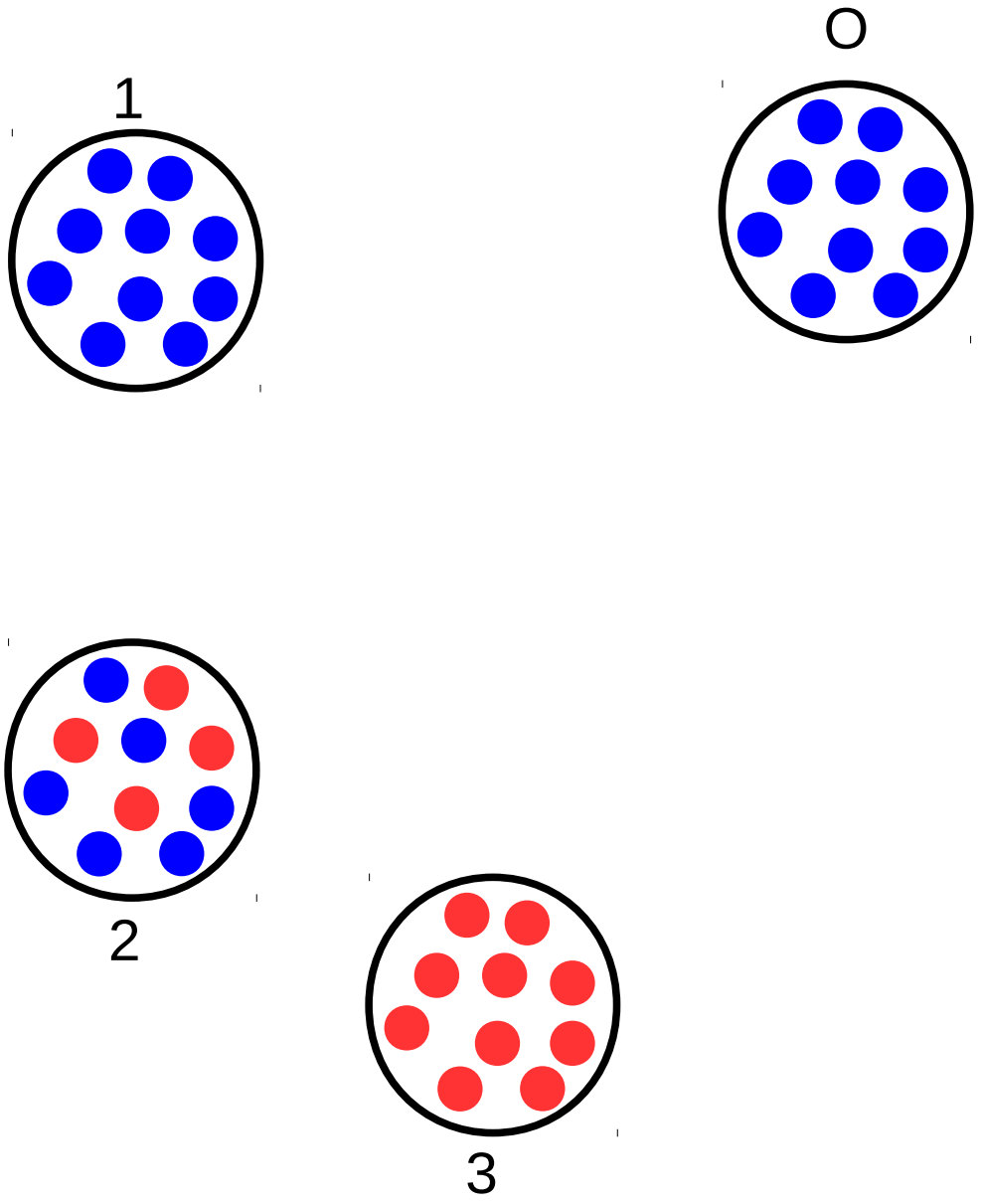

O
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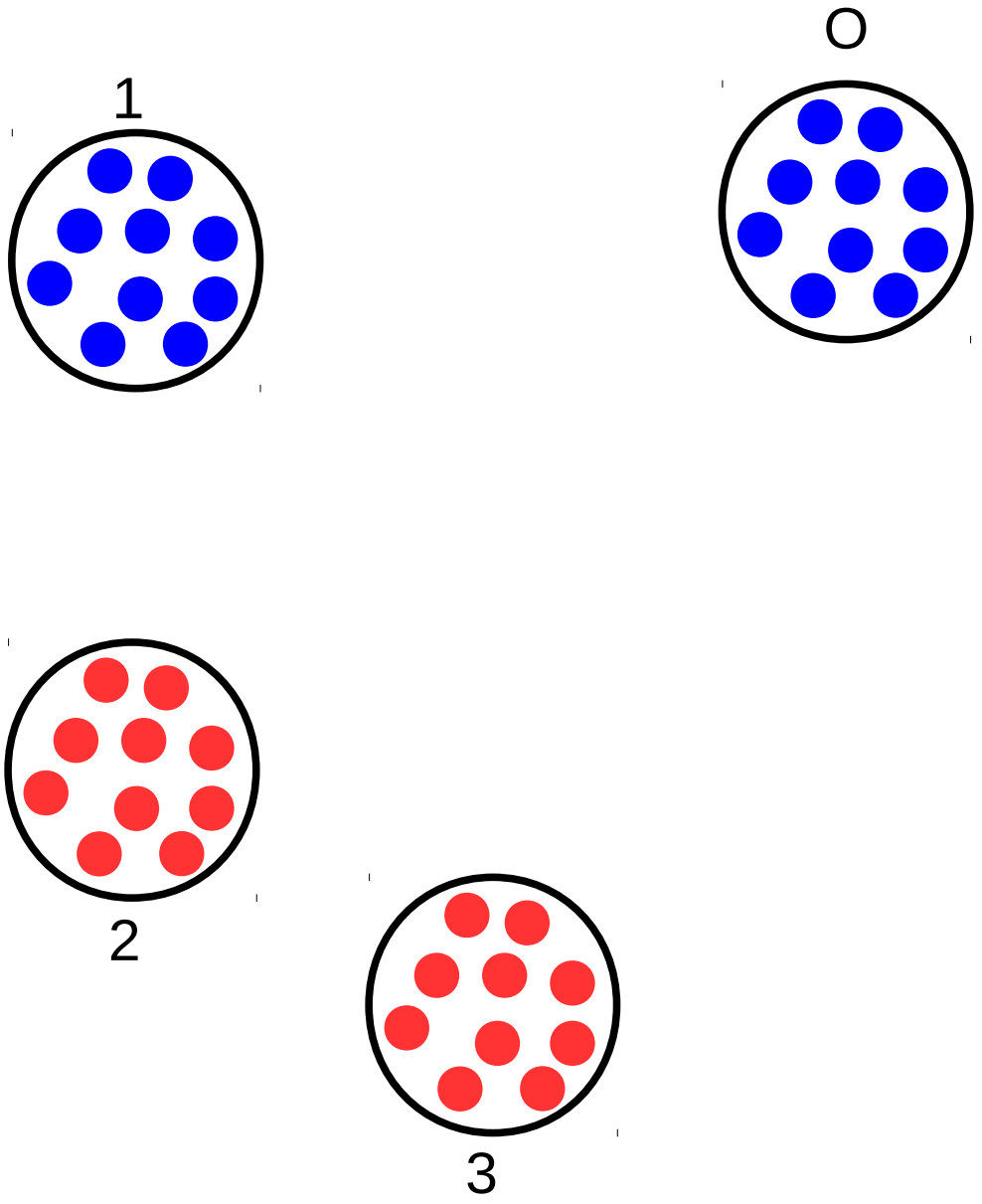

O
1
2
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# Gene flow

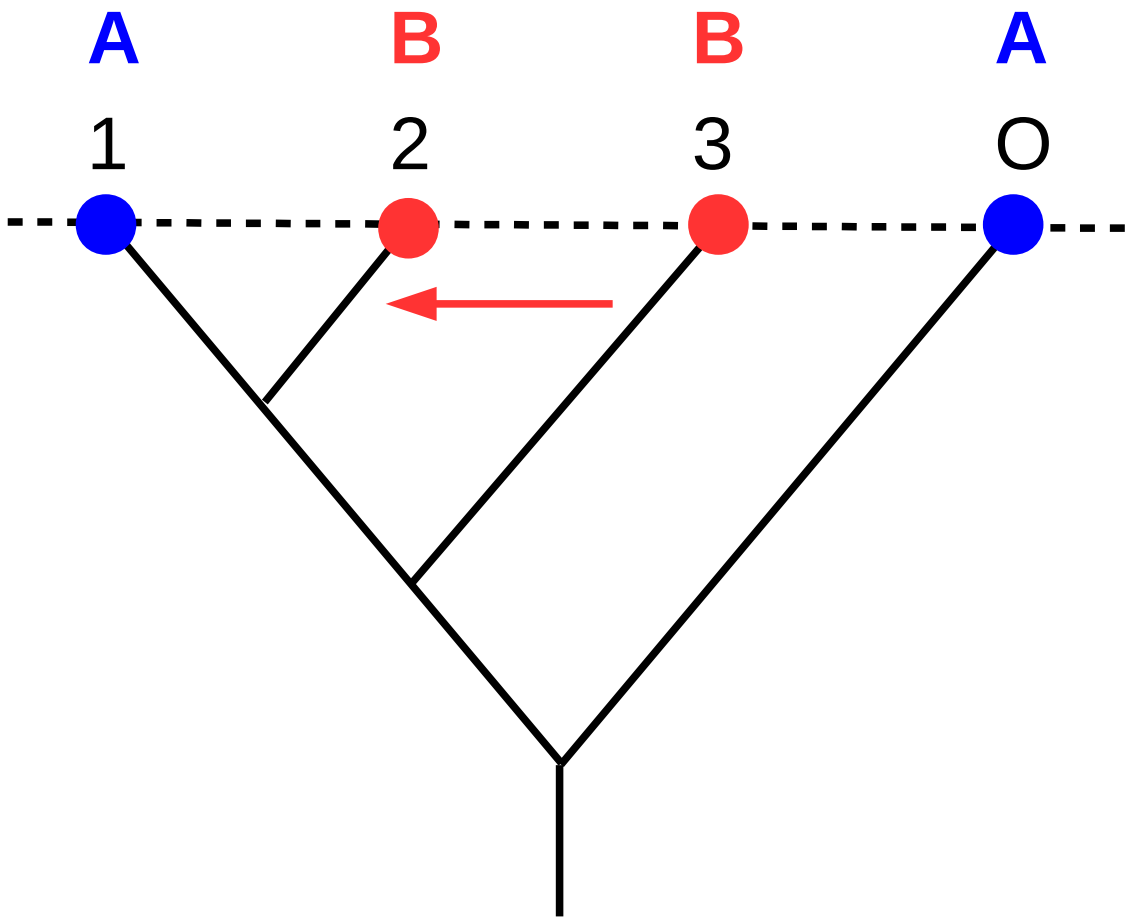



O
1
2
3


O
1
2
3


1
O
2
3

# Gene flow

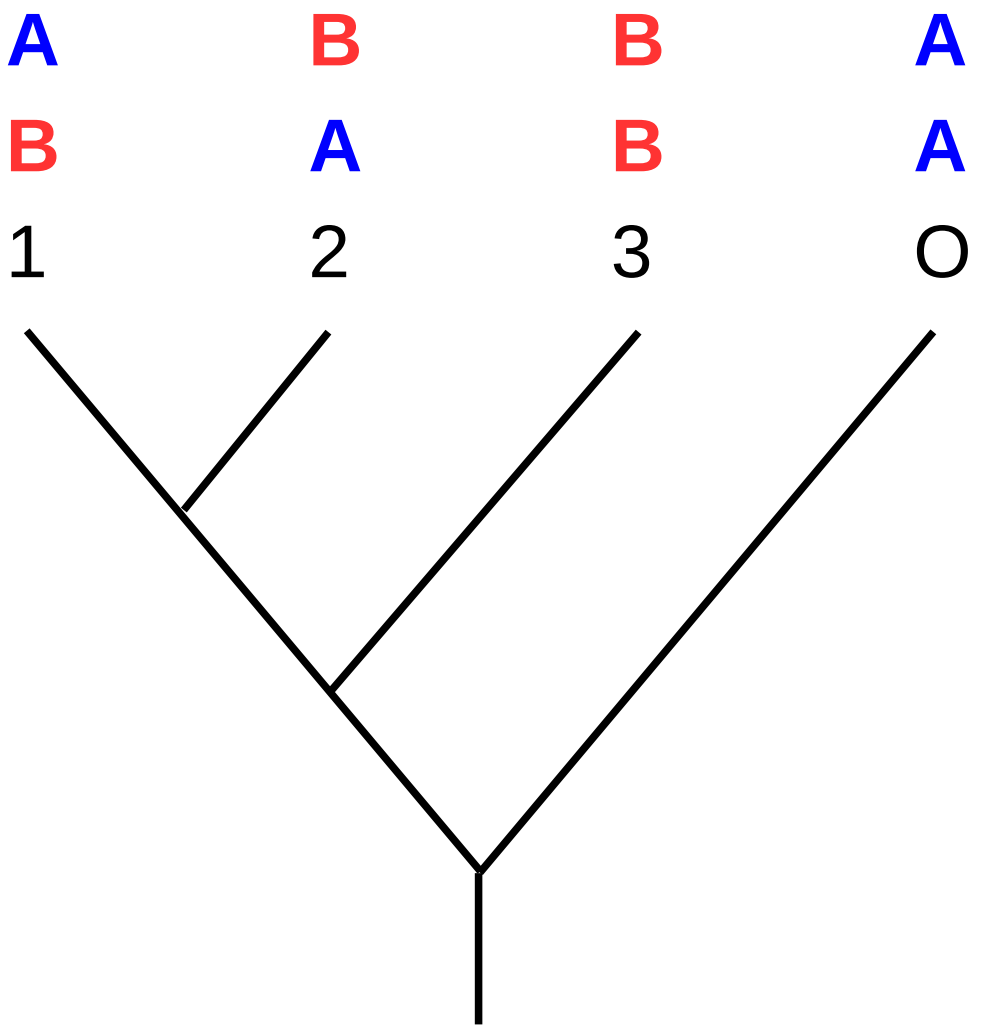



**Gene tree does not = species tree**

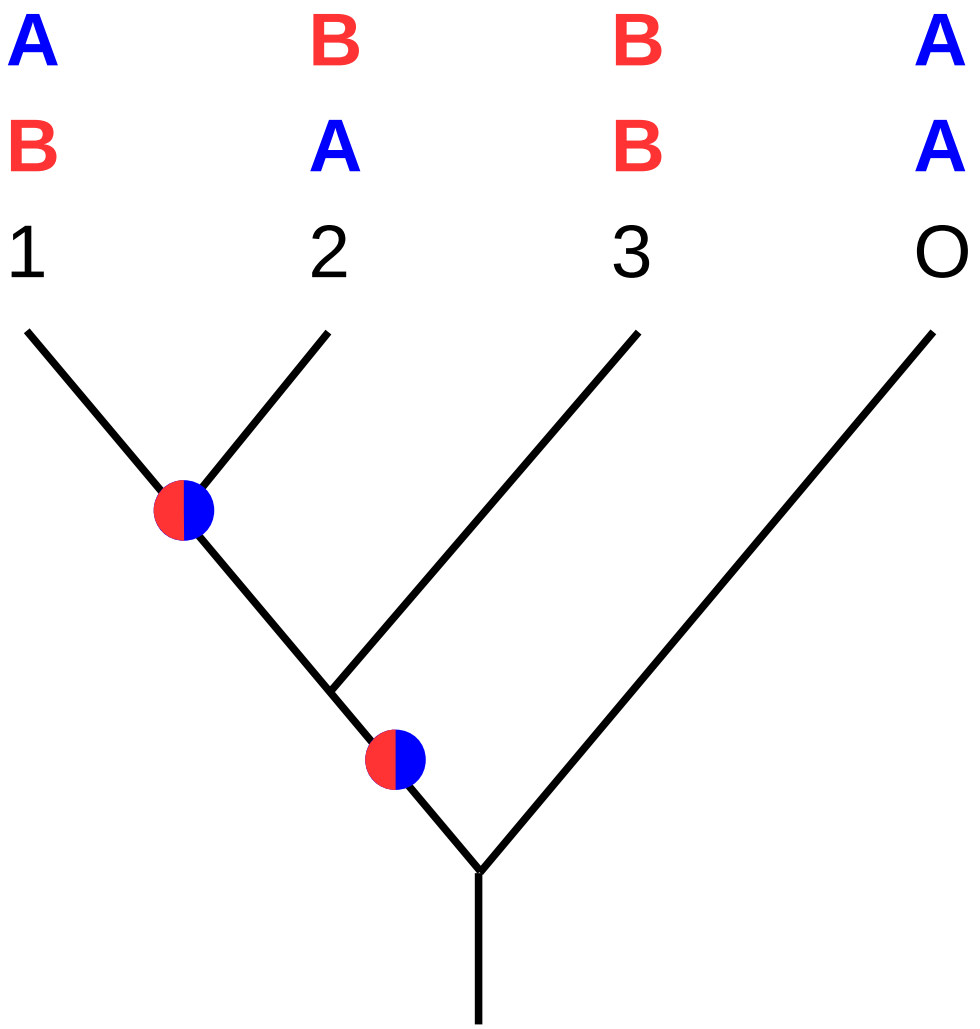
# Gene flow

- Gene flow may also produce gene trees that are incongruent with the species tree
- Basically the same pattern as ILS
- How do we differentiate gene flow and ILS?

# Expected frequencies of incongruent gene trees

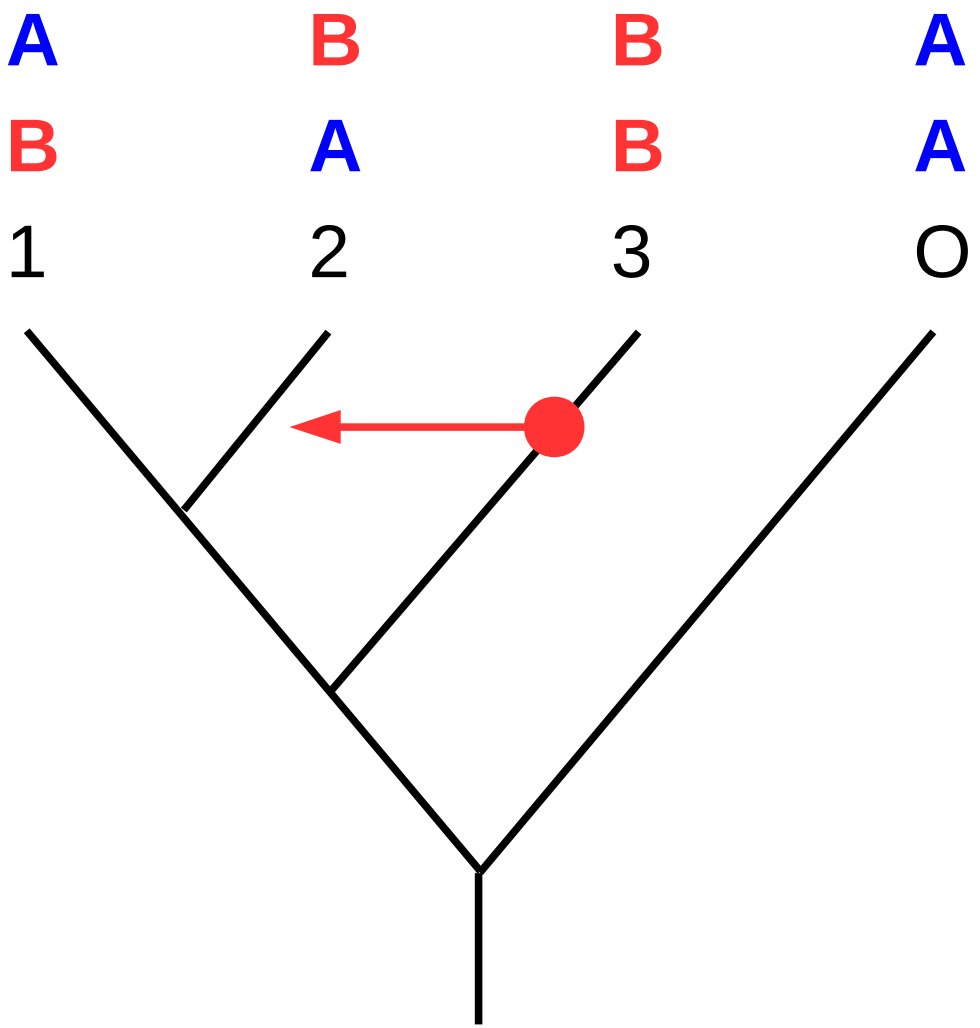


# Expected frequencies of incongruent gene trees




**Incomplete lineage sorting?**

# Expected frequencies of incongruent gene trees




**Gene flow?**

# Test of admixture – D statistic

- If gene tree / species tree incongruence results solely from ILS, ABBA and BABA should occur at equal frequencies

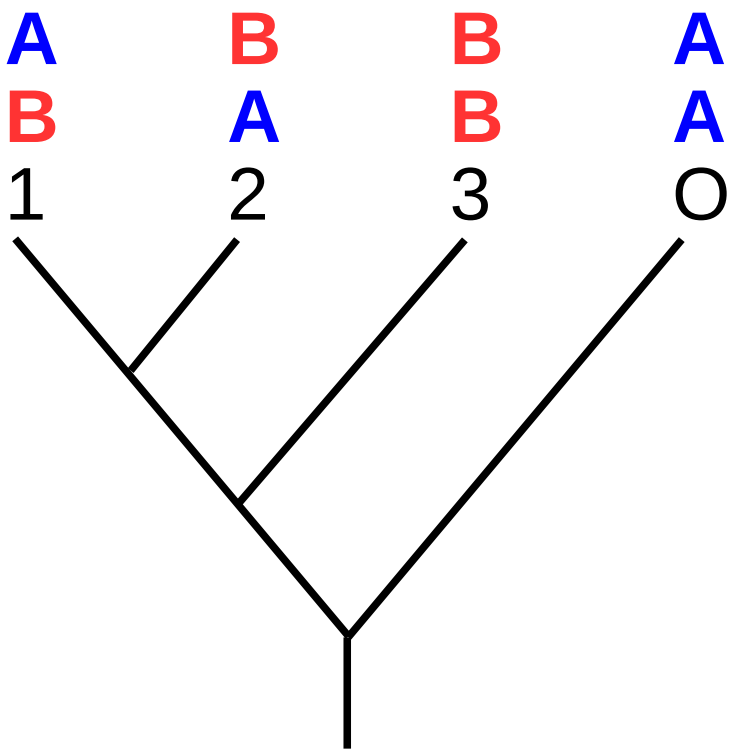
# Test of admixture – D statistic

- If gene tree / species tree incongruence results solely from ILS, ABBA and BABA should occur at equal frequencies
- If ABBA and BABA occur at different frequencies, then we must invoke admixture

# Test of admixture – D statistic

- If gene tree / species tree incongruence results solely from ILS, ABBA and BABA should occur at equal frequencies
- If ABBA and BABA occur at different frequencies, then we must invoke admixture

$$D = \frac{\mathbf{ABBA - BABA}}{\mathbf{ABBA + BABA}}$$

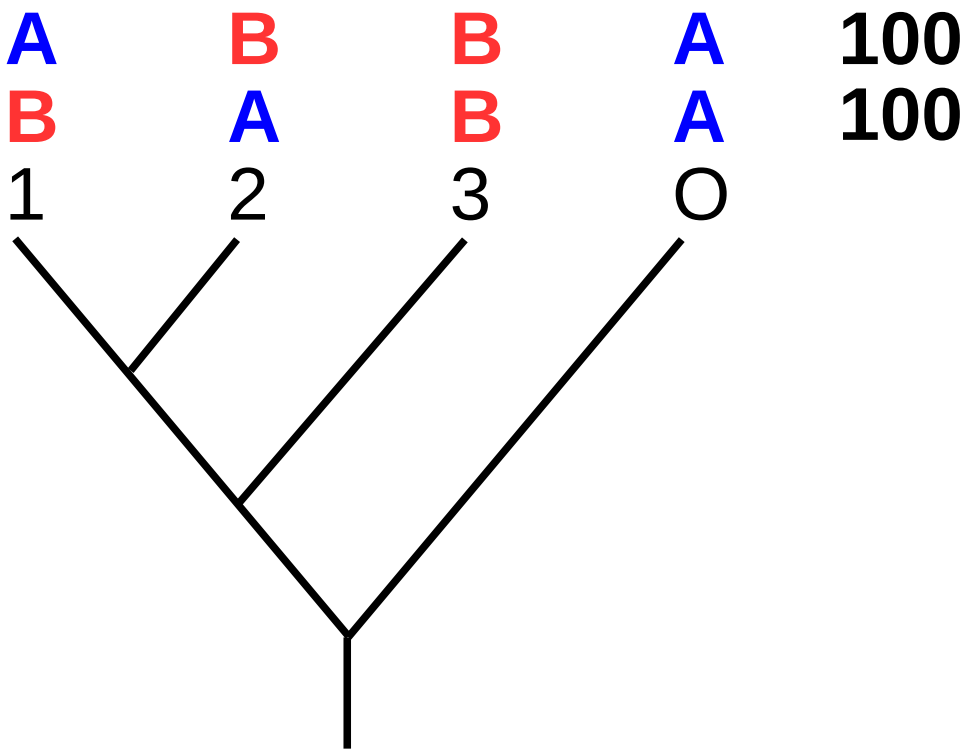



1 = individual 1

2 = individual 2

3 = candidate admixing species/population

O = outgroup

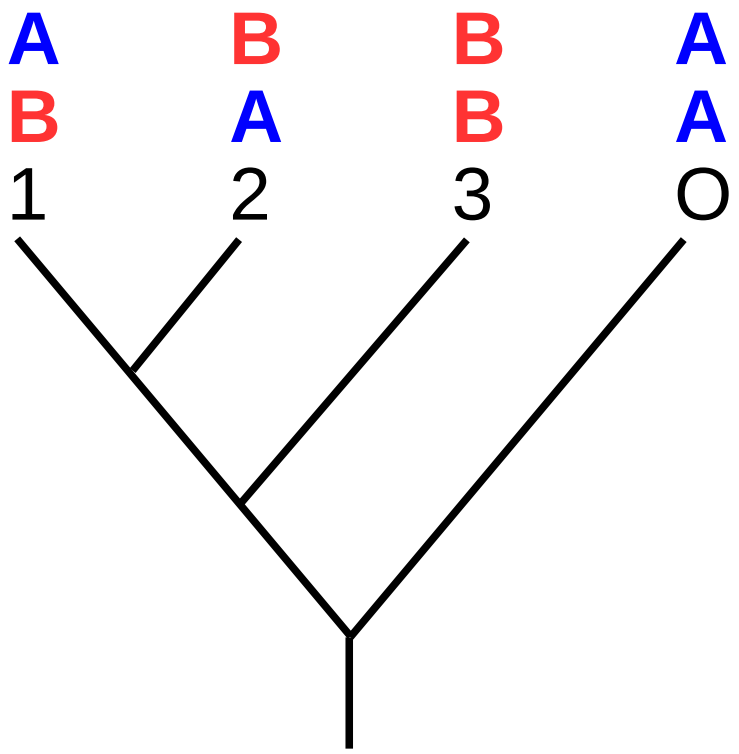



1 = individual 1

2 = individual 2

3 = candidate admixing species/population

O = outgroup




$$D = \frac{ABBA - BABA}{ABBA + BABA}$$

$$D = \frac{100 - 100}{100 + 100}$$

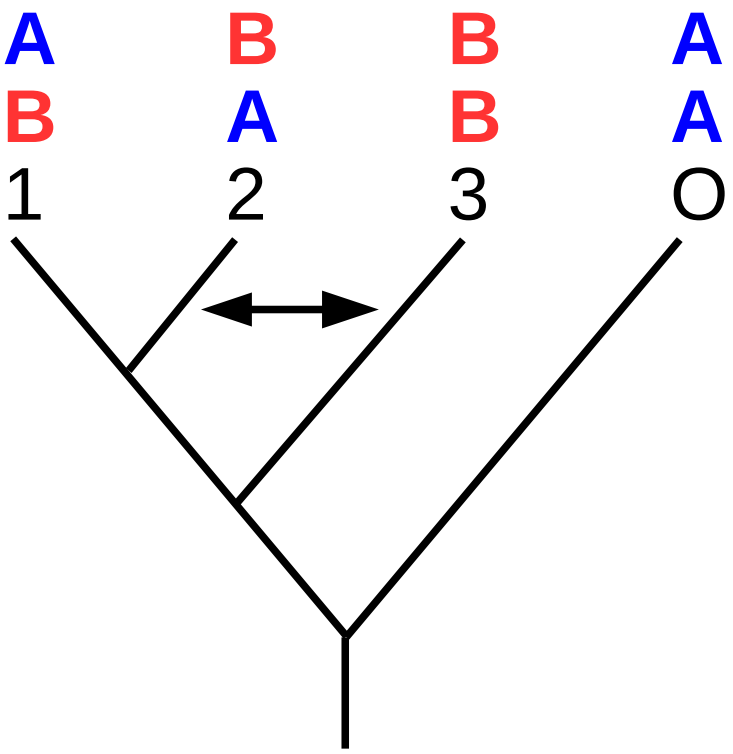
$$D = 0$$

1 = individual 1

2 = individual 2

3 = candidate admixing species/population

O = outgroup




$$D = \frac{ABBA - BABA}{ABBA + BABA}$$

$$D = \frac{200 - 100}{200 + 100}$$

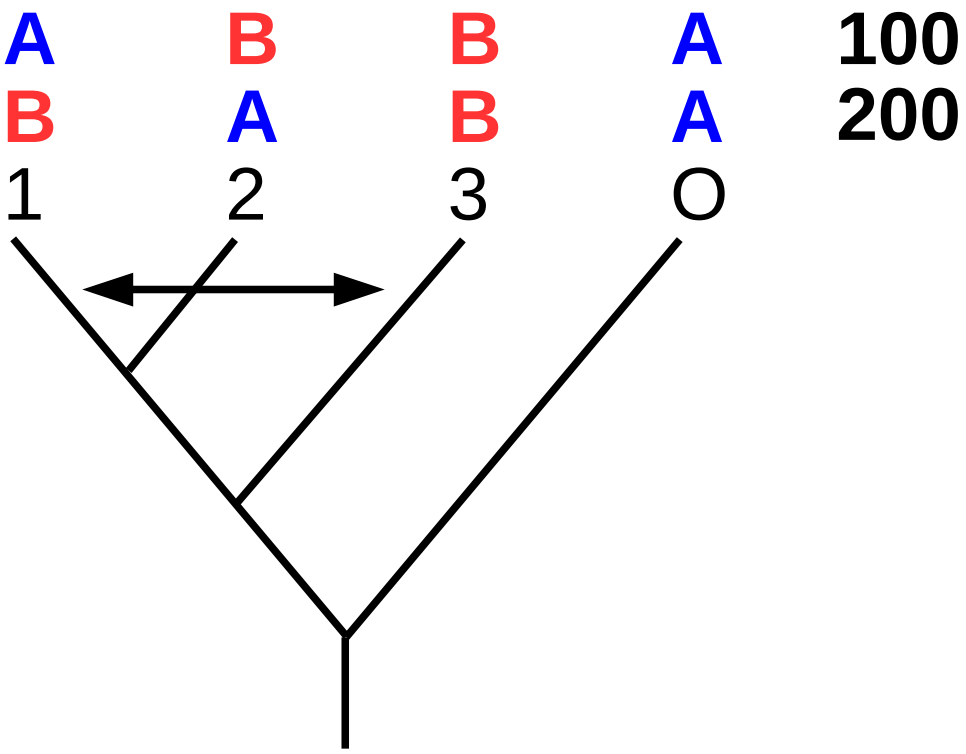
$$D = 0.33$$

1 = individual 1

2 = individual 2

3 = candidate admixing species/population

O = outgroup




$$D = \frac{ABBA - BABA}{ABBA + BABA}$$

$$D = \frac{100 - 200}{100 + 200}$$

$$D = -0.33$$

1 = individual 1

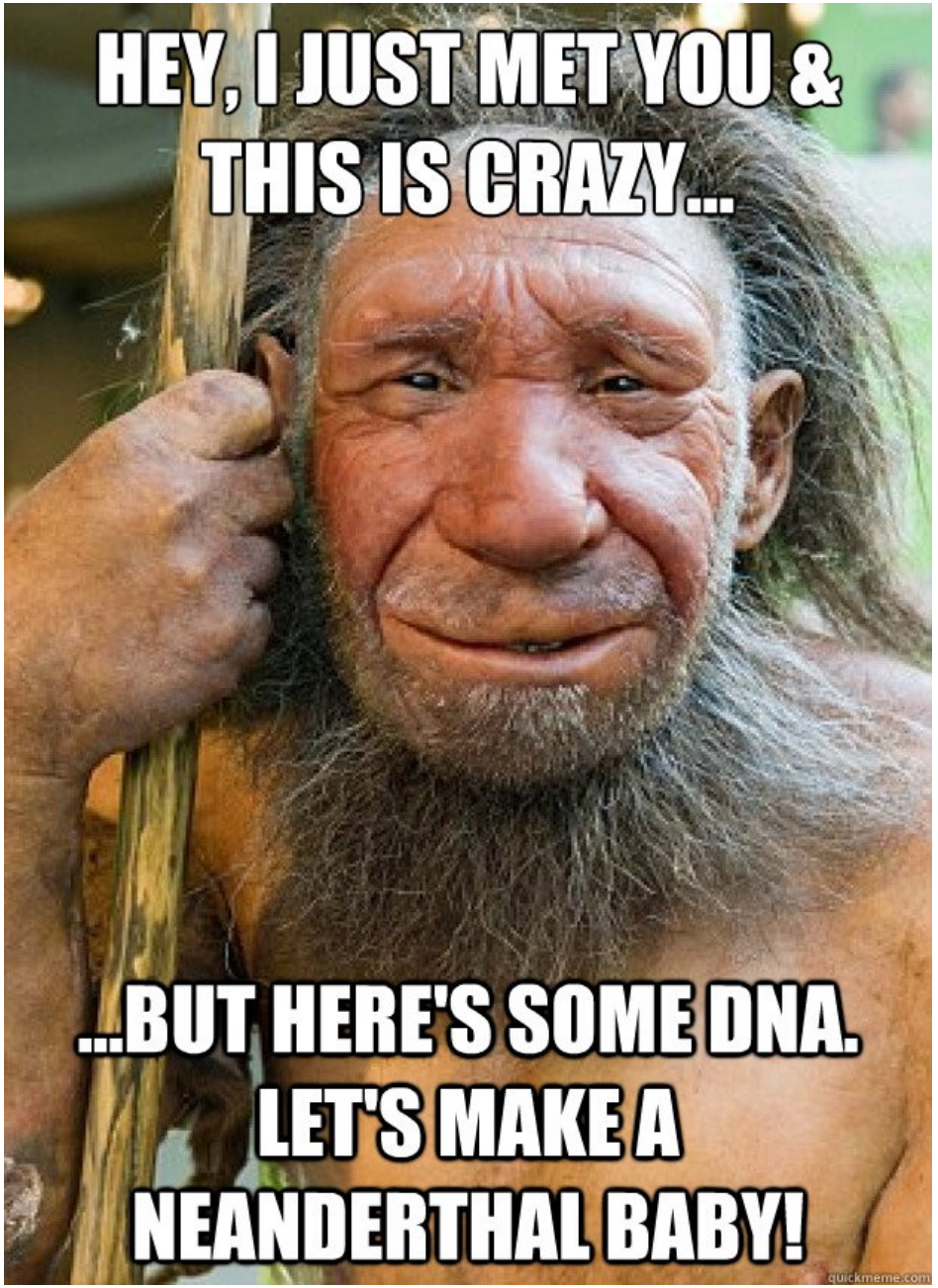
2 = individual 2

3 = candidate admixing species/population

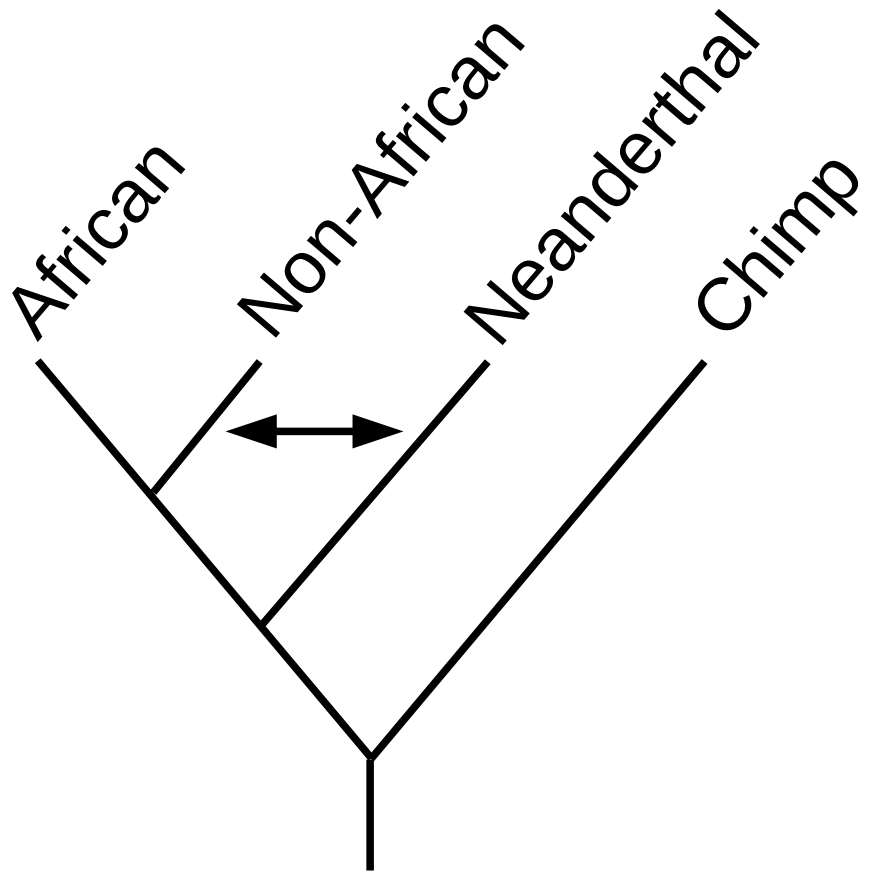
O = outgroup

# Test of admixture – D statistic

- *D* ranges between -1 and +1
- *D* = 0 in the absence of admixture (or equal admixture)
- + or – identifies the (more) admixed individual
- The test is relative (differential admixture between individuals)
- The direction of gene flow remains unknown (but other tests exist)
- Meaningful results require large numbers of ABBA and BABA sites
- Whole genome data


HEY, I JUST MET YOU &
THIS IS CRAZY...
...BUT HERE'S SOME DNA.
LET'S MAKE A
NEANDERTHAL BABY!
quickmeme.com

# Human – Neanderthal admixture




| P1 | P2 | *D* |
|---|---|---|
| African | African | -0.001 (n.s.) |
| African | French | 0.04* |
| African | Papuan | 0.039* |
| African | Han | 0.05* |
| African | French | 0.045* |
| African | Papuan | 0.044* |
| African | Han | 0.053* |

Green, Richard E., et al. "A draft sequence of the Neandertal genome."Science328.5979 (2010): 710-722.